UNIVERSITAS INDONESIA

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FAKULTAS ILMU KOMPUTER

PENGANTAR STATISTIKA

Descriptive Statistics II: Frequency, Central Tendencies, Variability, & z-scores

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LEARNING OBJECTIVES

1. • Know the concept of frequency
2. • Know the concept of central tendency
3. • Know the concept of variability
4. • Know the concept of z-score

2.1 Frequency Distributions

- The job of descriptive statistics: to simplify the organization and presentation of data. One of the most common procedures for organizing a set of data is to place the scores in a *frequency distribution*
- A **frequency distribution** is
  - An organized tabulation
  - Showing the number of individuals located in each category on the scale of measurement
- Can be either a **table** or a **graph**
- Always shows
  - The categories that make up the scale
  - The frequency, or number of individuals, in each category

2.2 Frequency Distribution Tables

- Structure of Frequency Distribution Table
  - Categories in a column (often ordered from highest to lowest but could be reversed)
  - Frequency count next to category
- $\Sigma f = N$
- To compute ΣX from a table
  - Convert table back to original scores *or*
  - Compute ΣfX

X	*F*
5	1
4	2
3	3
2	3
1	1

X	*f*	*fX*	
5	1	5	(the one 5 totals 5)
4	2	8	(the two 4s total 8)
3	3	9	(the three 3s total 9)
2	3	6	(the three 2s total 6)
1	1	1	(the one 1 totals 1)
	$\Sigma X = 29$		

Proportions and Percentages

Proportions

- Measures the fraction of the total group that is associated with each score
- $proportion = p = \frac{f}{N}$
- Called ***relative frequencies*** because they describe the frequency (f) *in relation to the total number* (N)

Percentages

- Expresses relative frequency out of 100
- $percentage = p(100) = \frac{f}{N}(100)$
- Can be included as a separate column in a frequency distribution table

Example 2.3
Frequency, Proportion and Percent

X	f	p = f/N	percent = p(100)
5	1	1/10 = .10	10%
4	2	2/10 = .20	20%
3	3	3/10 = .30	30%
2	3	3/10 = .30	30%
1	1	1/10 = .10	10%

Learning Check

- Use the Frequency Distribution Table to determine how many subjects were in the study

A • 10

B • 15

C • 33

D • Impossible to determine

X	f
5	2
4	4
3	1
2	0
1	3

Learning Check - Answer

- Use the Frequency Distribution Table to determine how many subjects were in the study

A • **10**

B • 15

C • 33

D • Impossible to determine

X	f
5	2
4	4
3	1
2	0
1	3

Learning Check

- For the frequency distribution shown, is each of these statements True or False?

X	f
5	2
4	4
3	1
2	0
1	3

T/F

- More than 50% of the individuals scored above 3

T/F

- The proportion of scores in the lowest category was $p = 3$

Learning Check - Answer

- For the frequency distribution shown, is each of these statements True or False?

X	f
5	2
4	4
3	1
2	0
1	3

True

- Six out of ten individuals scored above 3 = 60% = more than half

False

- A proportion is a fractional part; 3 out of 10 scores = 3/10 = .3

Grouped Frequency Distribution Tables

- If the number of categories is very large, they are combined (grouped) to make the table easier to understand
- However, information is lost when categories are grouped
  - Individual scores cannot be retrieved
  - The wider the grouping interval, the more information is lost

"Rules" for Constructing Grouped Frequency Distributions

- Requirements (Mandatory Guidelines)
  - All intervals must be the same width
  - Make the bottom (low) score in each interval a multiple of the interval width
- "Rules of Thumb" (Suggested Guidelines)
  - Ten or fewer class intervals is typical (but use good judgment for the specific situation)
  - Choose a "simple" number for interval width

Discrete Variables in Frequency *or* Grouped Distributions

- Constructing either frequency distributions or grouped frequency distributions for ***discrete variables*** is uncomplicated
  - Individuals with the same recorded score had precisely the same measurements
  - The score is an ***exact*** score

Continuous Variables in Frequency Distributions

- Constructing frequency distributions for ***continuous*** variables requires understanding that a score actually represents an interval
  - A given "score" actually could have been any value within the score's ***real limits***
  - The recorded value was rounded off to the middle value between the score's real limits
  - Individuals with the same ***recorded score*** probably differed slightly in their actual performance

Continuous Variables in Frequency Distributions

- Constructing grouped frequency distributions for ***continuous variables*** also requires understanding that a score actually represents an interval
- Consequently, grouping several scores actually requires grouping several intervals
- Apparent limits of the (grouped) class interval are always ***one unit*** smaller than the real limits of the (grouped) class interval. (Why?)

2.3 Frequency Distribution Graphs

- Pictures of the data organized in tables
  - All have two axes
  - X-axis (abscissa) typically has categories of measurement scale increasing left to right
  - Y-axis (ordinate) typically has frequencies increasing bottom to top
- General principles
  - Both axes should have value 0 where they meet
  - Height should be about ⅔ to ¾ of length

Data Graphing Questions

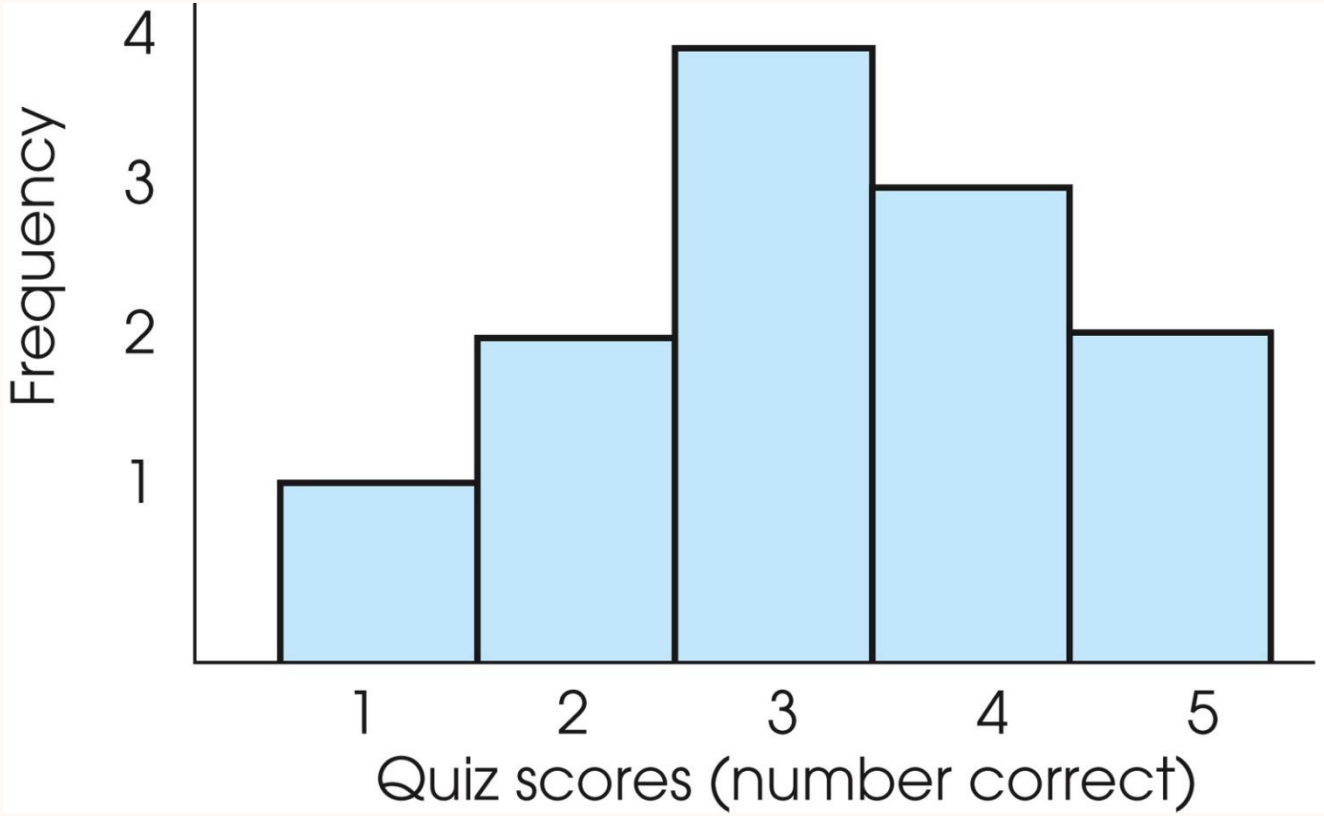
- Level of measurement? (nominal; ordinal; interval; or ratio)
- Discrete or continuous data?
- Describing samples or populations?

The answers to these questions determine which is the appropriate graph

Frequency Distribution Histogram

- Requires numeric scores (interval or ratio)
- Represent ***all*** scores on X-axis from minimum thru maximum observed data values
- Include ***all*** scores with frequency of zero
- Draw bars above each score (interval)
  - Height of bar corresponds to frequency
  - Width of bar corresponds to score real limits (or one-half score unit above/below discrete scores)

Frequency Distribution Histogram



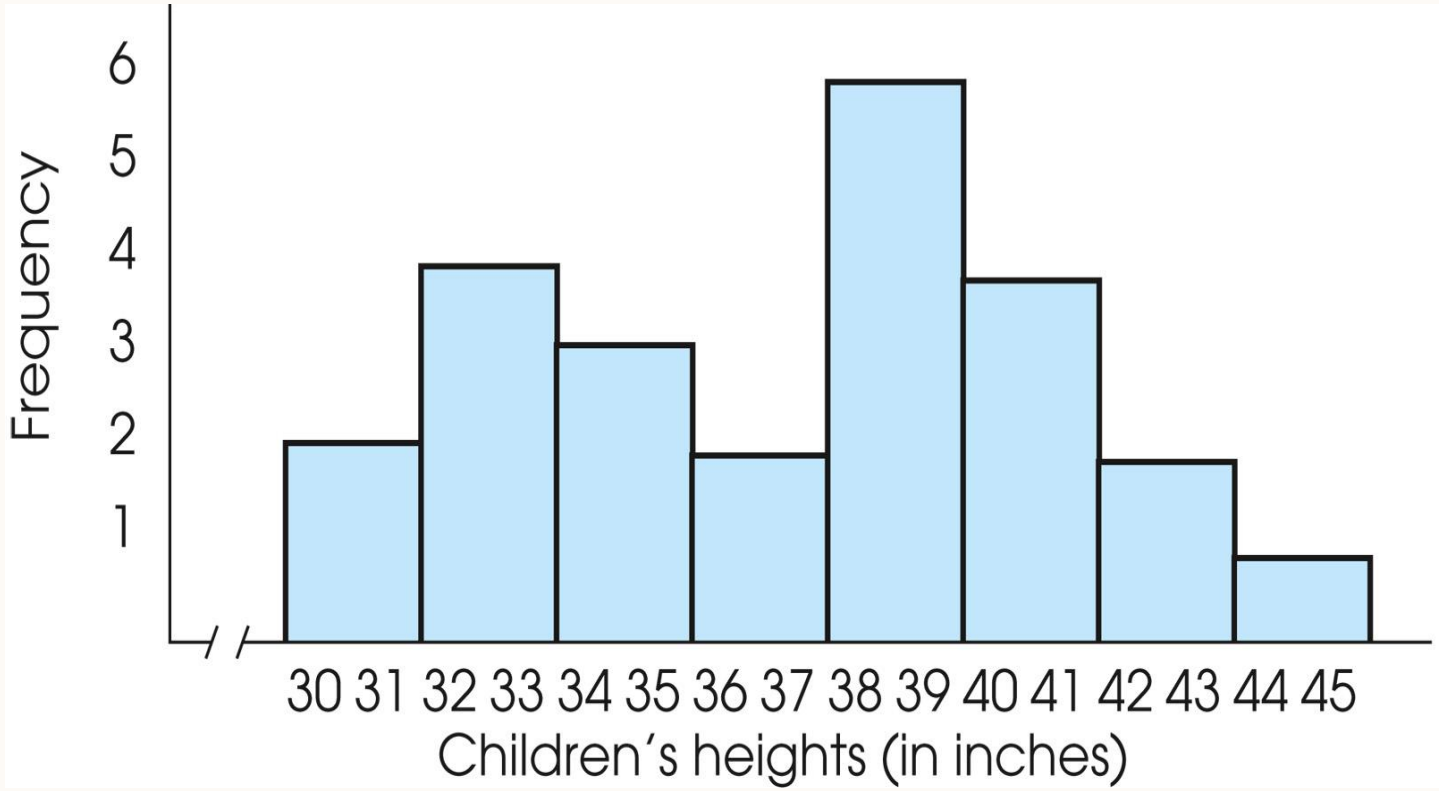
X	f
5	2
4	3
3	4
2	2
1	1

Grouped Frequency Distribution Histogram

Same requirements as for frequency distribution histogram ***except:***

- Draw bars above each (grouped) class interval
  - Bar width is the *class interval* real limits
  - Consequence? Apparent limits are extended out one-half score unit at each end of the interval

Grouped Frequency Distribution

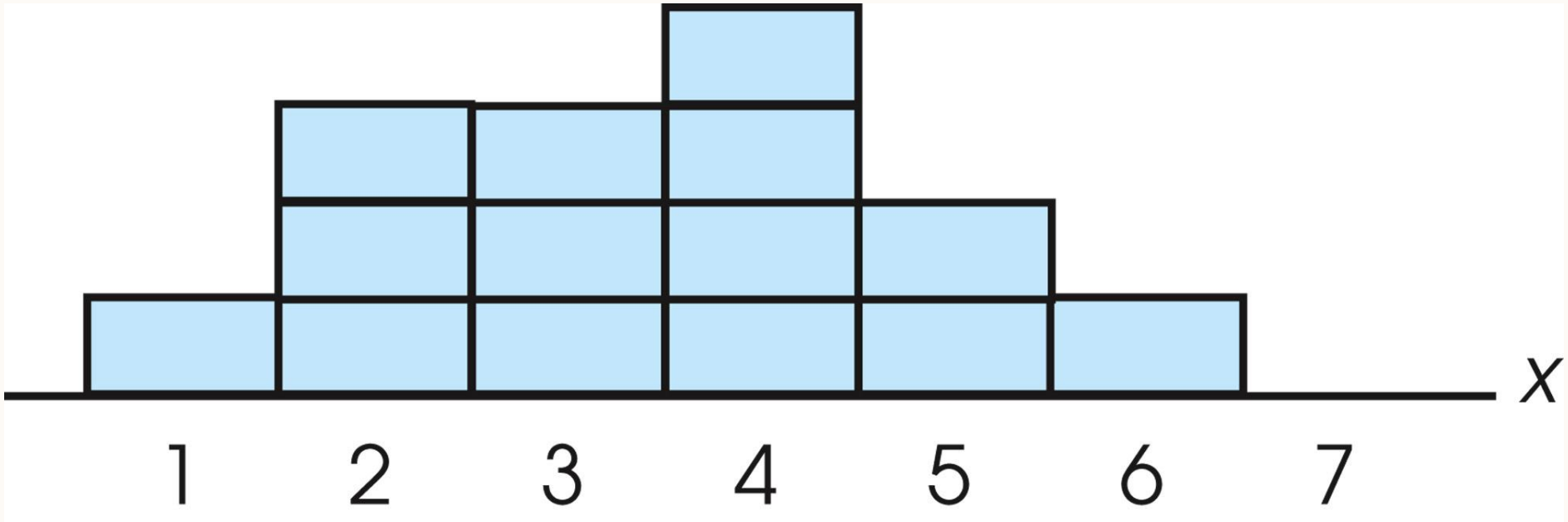


X	f
44–45	1
42–43	2
40–41	4
38–39	6
36–37	2
34–35	3
32–33	4
30–31	2

Block Histogram

- A histogram can be made a “block” histogram
- Create a bar of the correct height by drawing a stack of blocks
- Each block represents one per case
- Therefore, block histograms show the frequency count in each bar

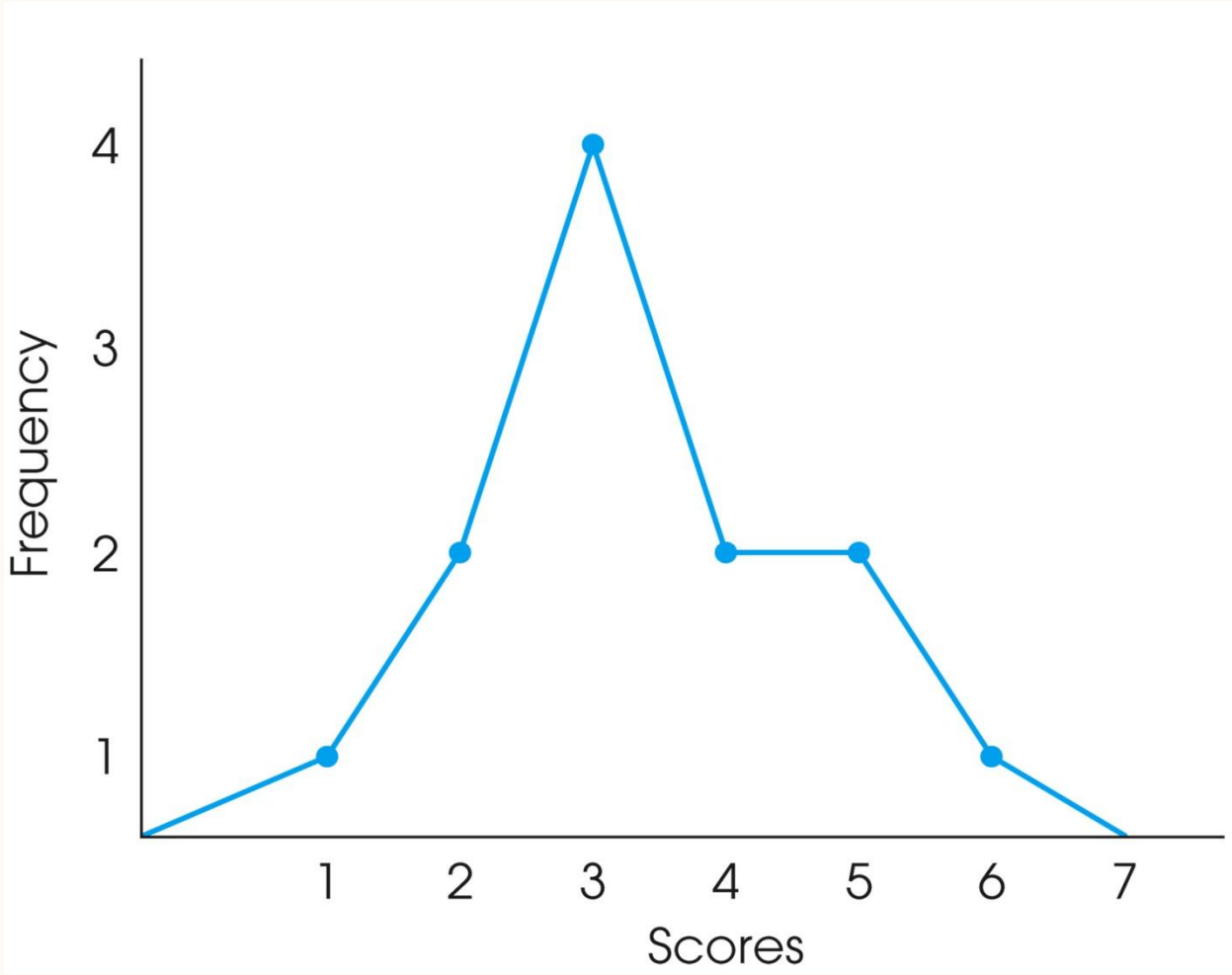
Frequency Distribution Block



Frequency Distribution Polygons

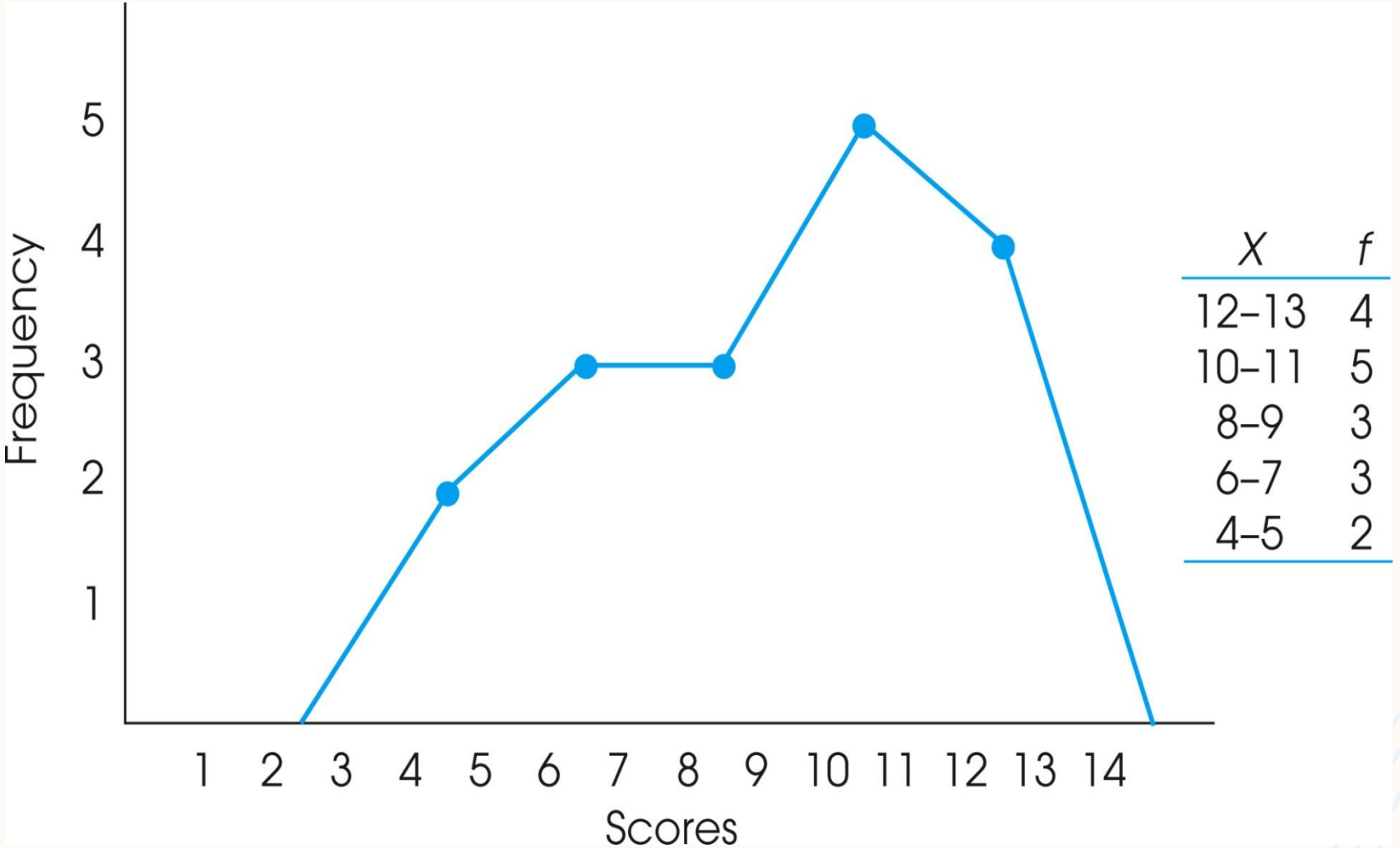
- List all numeric scores on the X-axis
  - Include those with a frequency of $f = 0$
- Draw a dot above the center of each interval
  - Height of dot corresponds to frequency
  - Connect the dots with a continuous line
  - Close the polygon with lines to the Y = 0 point
- Can also be used with grouped frequency distribution data

Frequency Distribution



X	f
6	1
5	2
4	2
3	4
2	2
1	1

Grouped Data Frequency Distribution

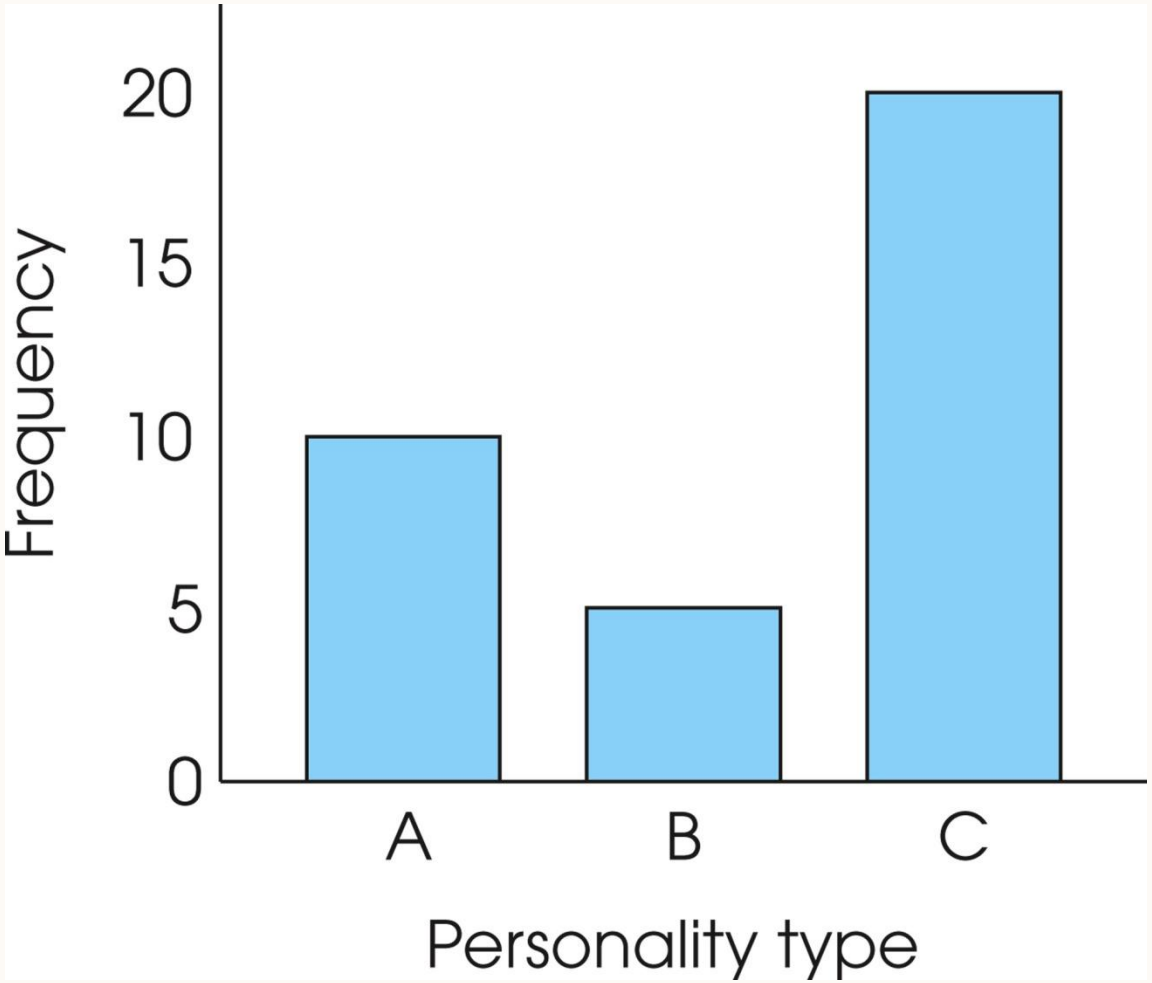


X	f
12–13	4
10–11	5
8–9	3
6–7	3
4–5	2

Graphs for Nominal or Ordinal Data

- For non-numerical scores (nominal and ordinal data), use a **bar graph**
  - Similar to a histogram
  - Spaces between adjacent bars indicates discrete categories
    - without a particular order (nominal)
    - non-measurable width (ordinal)

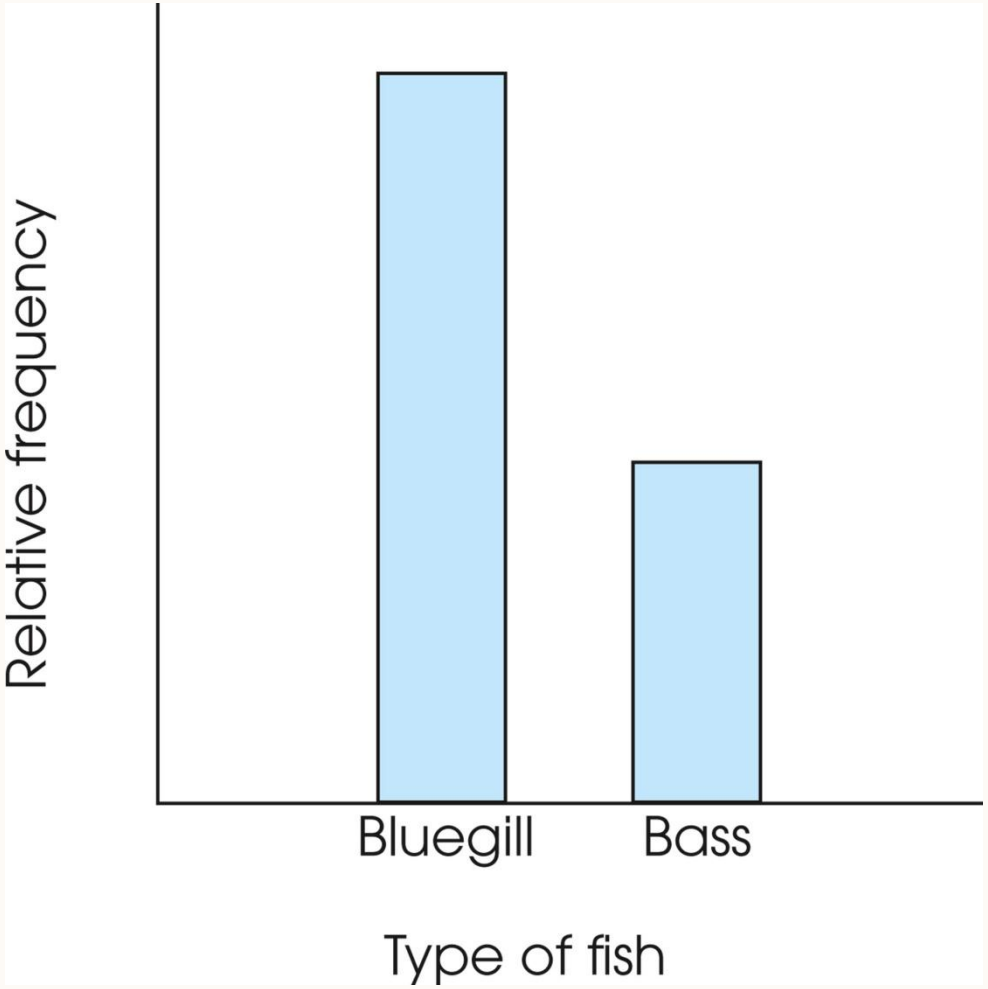
Figure 2.6 - Bar graph



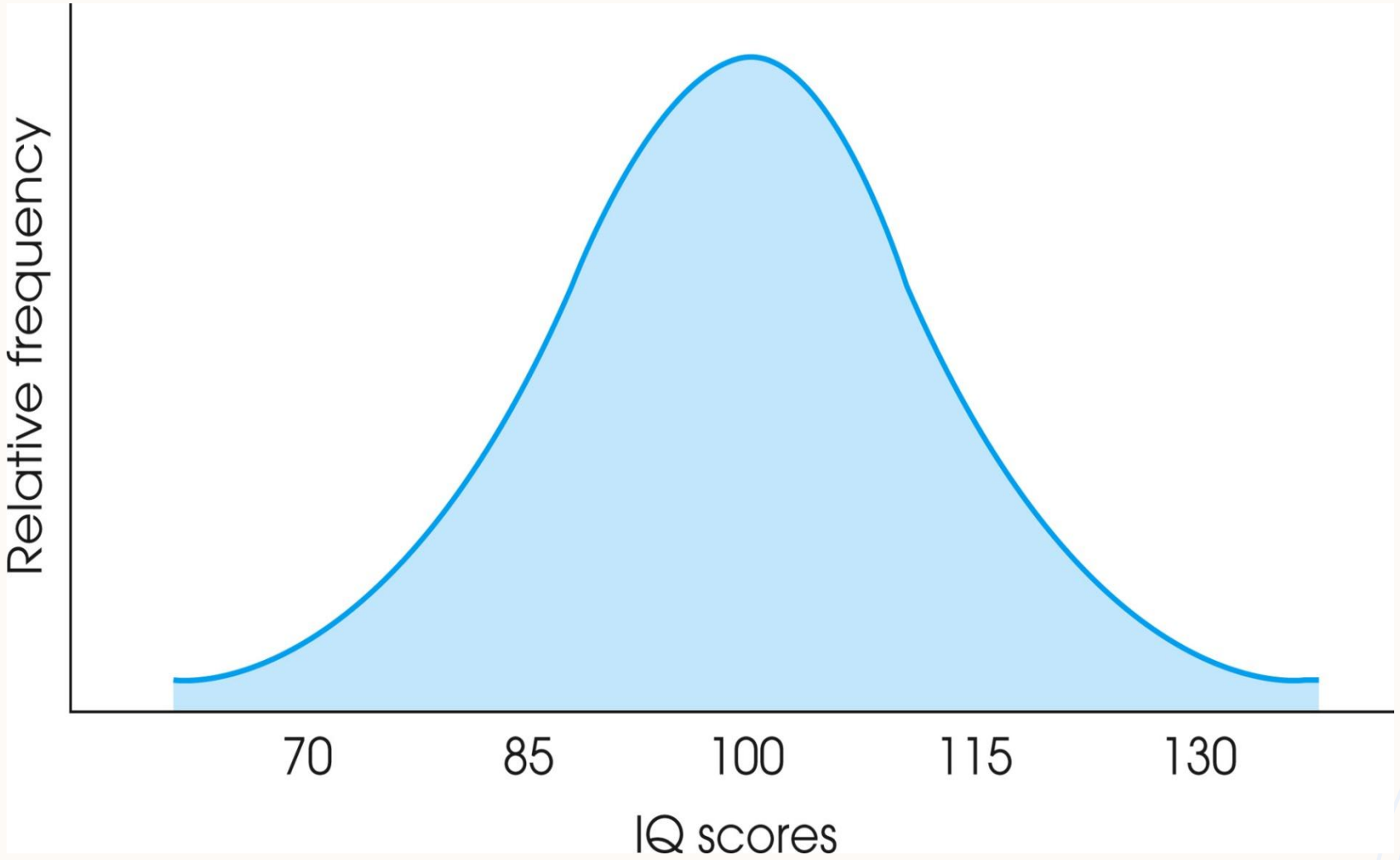
Population Distribution Graphs

- When population is small, scores for each member are used to make a histogram
- When population is large, scores for each member are not possible
  - Graphs based on relative frequency are used
  - Graphs use smooth curves to indicate exact scores were not used
- Normal
  - Symmetric with greatest frequency in the middle
  - Common structure in data for many variables

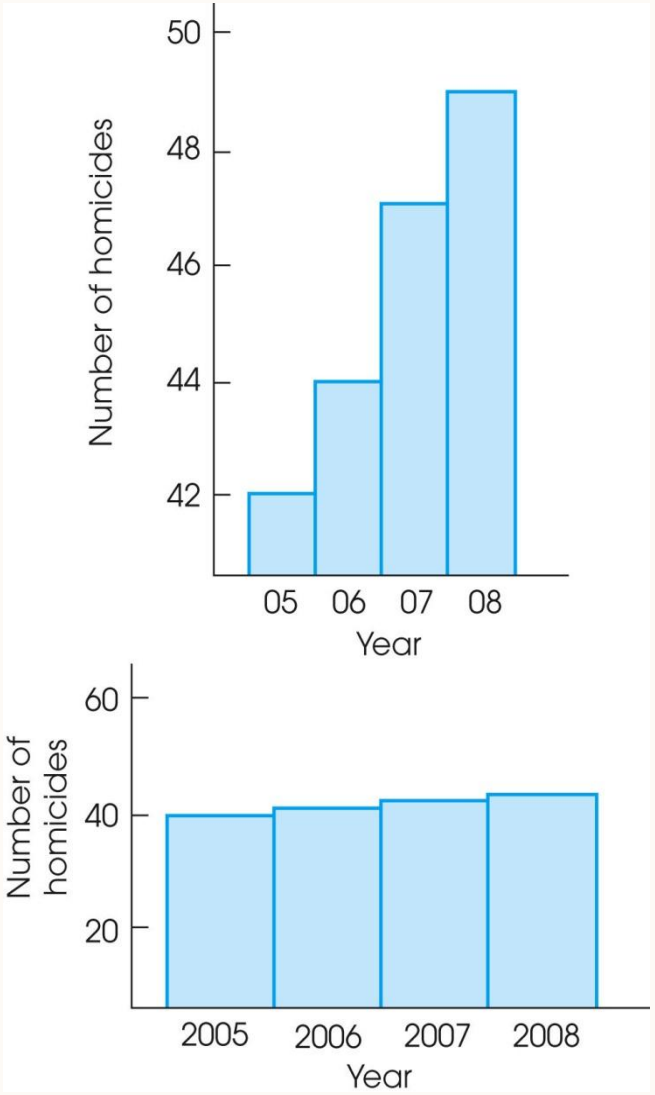
Bar Graph of Relative



Distribution Shown as a Normal Curve



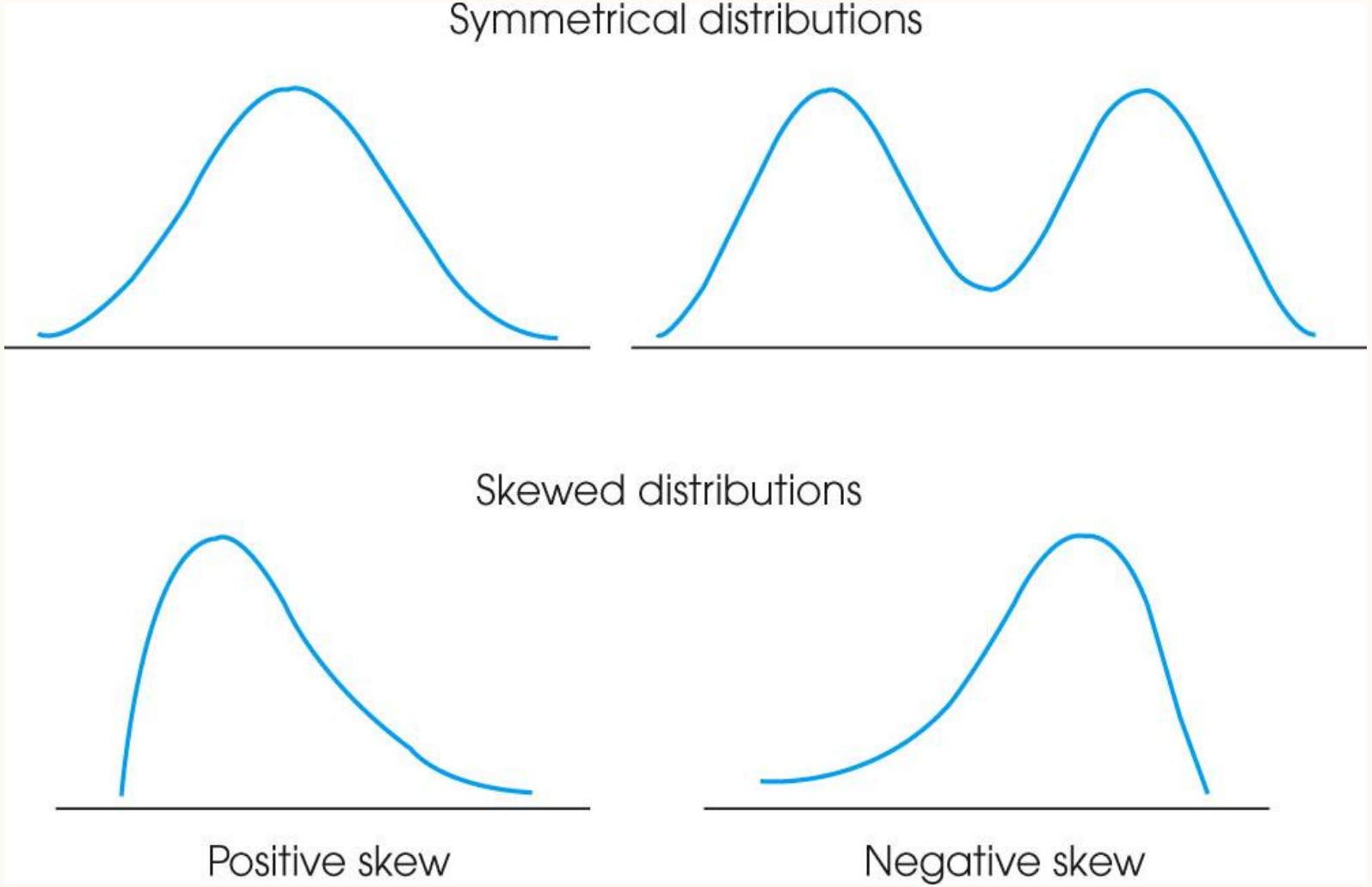
Use and Misuse of Graphs



2.4 Frequency Distribution Shape

- Researchers describe a distribution's shape in words rather than drawing it
- Symmetrical distribution: each side is a mirror image of the other
- Skewed distribution: scores pile up on one side and taper off in a **tail** on the other
  - Tail on the right (high scores) = positive skew
  - Tail on the left (low scores) = negative skew

Figure 2.10 - Distribution



Central Tendency

3.1 Defining Central Tendency

- Central tendency
  - A statistical measure
  - A single score to define the center of a distribution
- Purpose: find the single score that is most typical or best represents the entire group

Locate Each Distribution "Center"



Central Tendency Measures

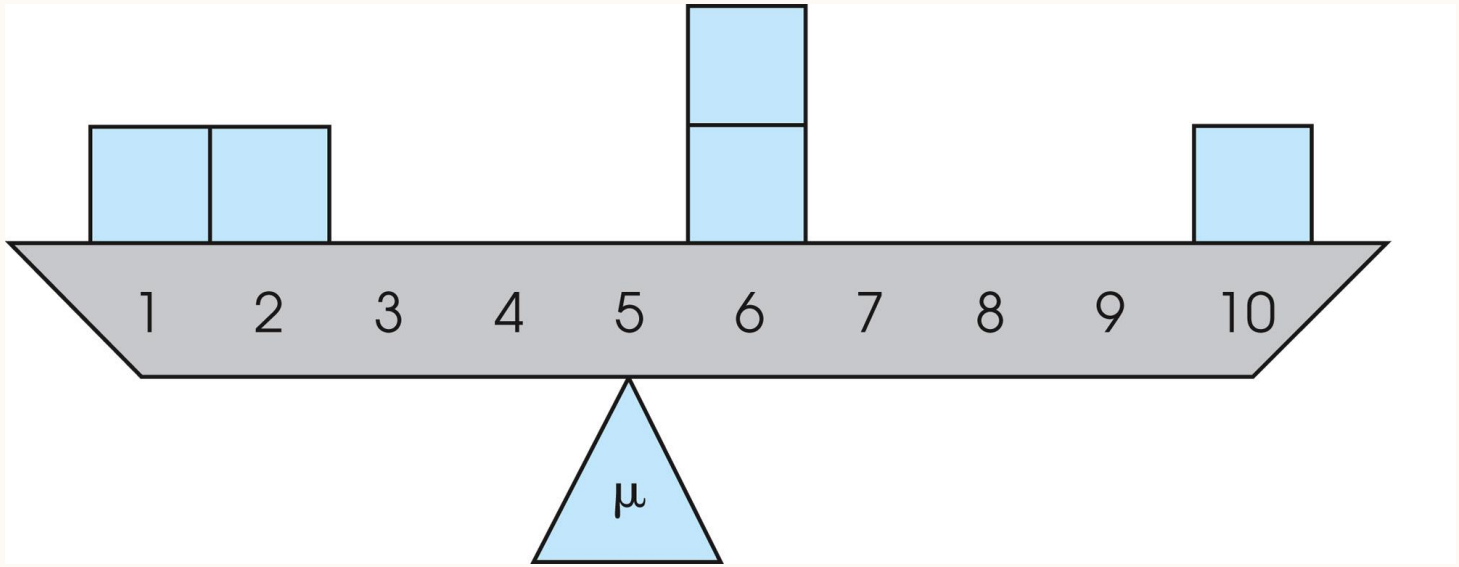
- Figure 3.1 shows that no single concept of central tendency is always the “best”
- Different distribution shapes require different conceptualizations of “center”
- Choose the one which best represents the scores in a specific situation

3.2 The Mean

- The mean is the sum of all the scores divided by the number of scores in the data.
- Population: $$\mu = \frac{\sum X}{N}$$
- Sample: $$M = \frac{\sum X}{n}$$

The Mean: Three Definitions

- Sum of the scores divided by the number of scores in the data
- The amount each individual receives when the total is divided equally among all the individuals in the distribution
- The balance point for the distribution



The Weighted Mean

- Combine two sets of scores
- Three steps:
  - Determine the combined sum of all the scores
  - Determine the combined number of scores
  - Divide the sum of scores by the total number of scores

Overall Mean = $$M = \frac{\sum X_1 + \sum X_2}{n_1 + n_2}$$

Computing the Mean from a Frequency Distribution Table

Table 3.1 (Modified)

Quiz Score (*X*)	*f*	*fX*
10	1	10
9	2	18
8	4	32
7	0	0
6	1	6
Total	$n = \Sigma f = 8$	$\Sigma fX = 66$
$M = \Sigma X / n = 66/8 = 8.25$		

Characteristics of the Mean

- Changing the value of a score changes the mean
- Introducing a new score or removing a score *changes* the mean (unless the score added or removed is *exactly* equal to the mean)
- Adding or subtracting a constant from each score changes the mean by the same constant
- Multiplying or dividing each score by a constant multiplies or divides the mean by that constant

3.3 The Median

- The median is the midpoint of the scores in a distribution when they are listed in order from smallest to largest
- The median divides the scores into two groups of equal size

Example 3.5

Locating the Median (odd *n*)

- Put scores in order
- Identify the "middle" score to find median

3 5 8 10 11

"Middle" score is 8 so median = 8

Example 3.6

Locating the Median (even *n*)

- Put scores in order
- Average middle pair to find median

1 1 4 5 7 9

(4 + 5) / 2 = 4.5

The Precise Median for a Continuous Variable

- A continuous variable can be infinitely divided
- The precise median is located in the interval defined by the real limits of the value.
- It is necessary to determine the fraction of the interval needed to divide the distribution exactly in half.

$$\text{fraction} = \frac{\text{number needed to reach 50\%}}{\text{number in the interval}}$$

Figure 3.4 – Finding a Precise Median for a Continuous Variable

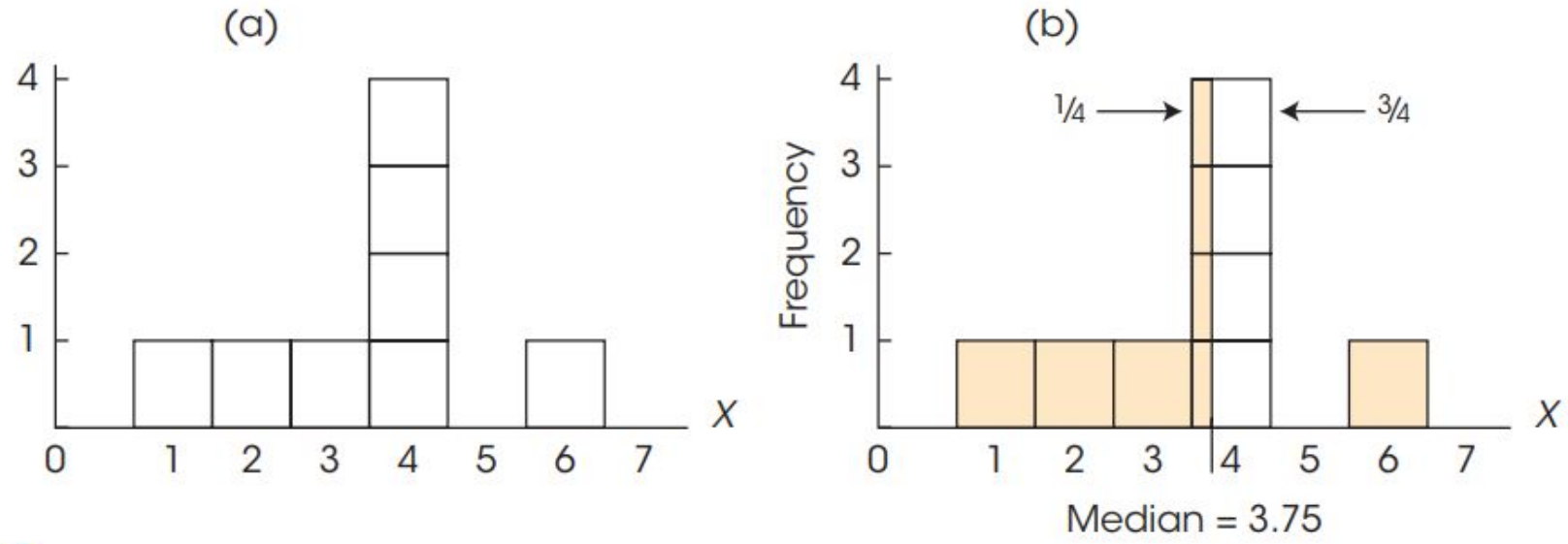


FIGURE 3.5

A distribution with several scores clustered at the median. The median for this distribution is positioned so that each of the four boxes at $X = 4$ is divided into two sections with $\frac{1}{4}$ of each box below the median (to the left) and $\frac{3}{4}$ of each box above the median (to the right). As a result, there are exactly 4 boxes, 50% of the distribution, on each side of the median.

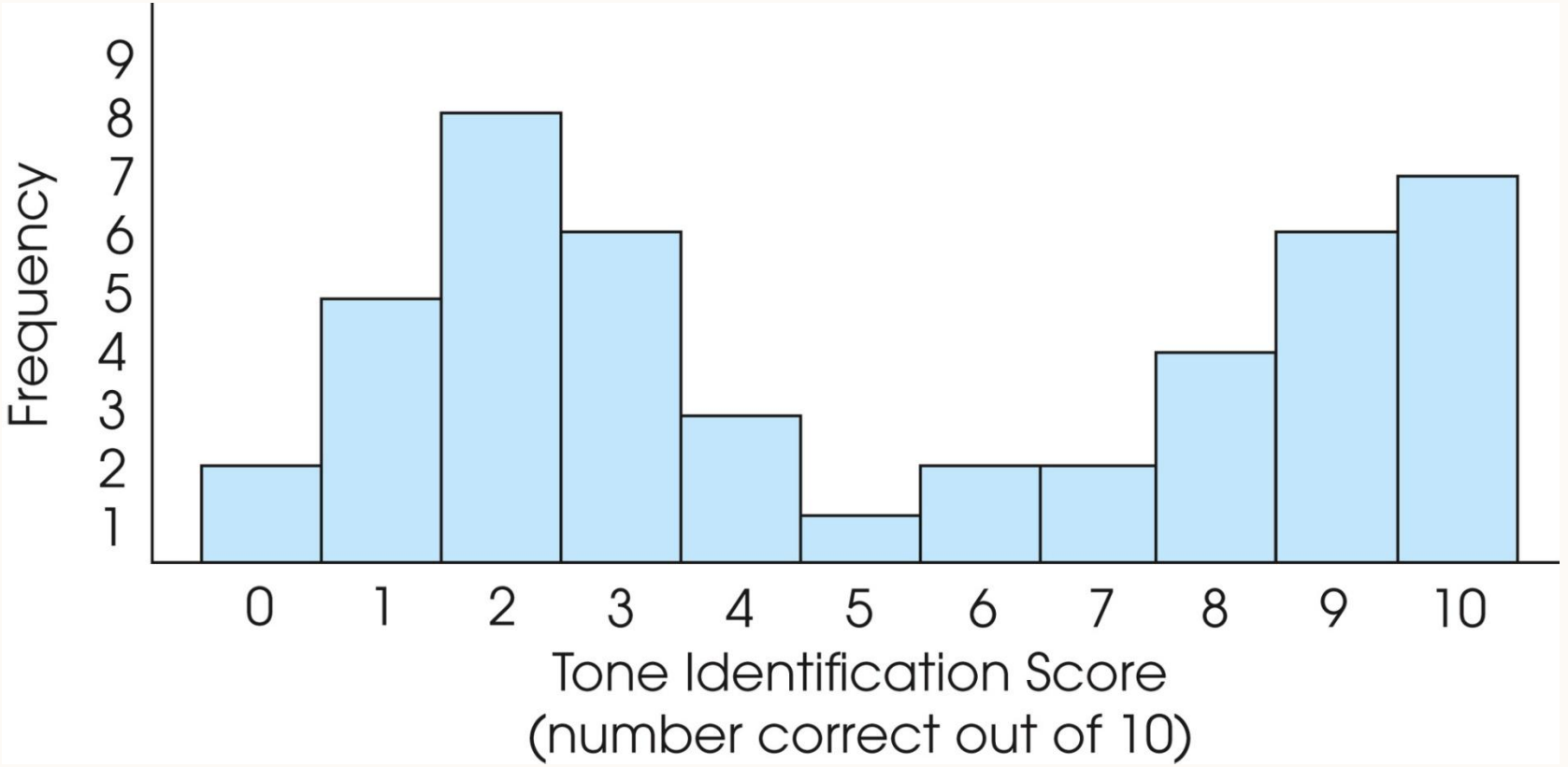
Median, Mean, and "Middle"

- Mean is the balance point of a distribution
  - Defined by distances
  - Often is not the midpoint of the scores
- Median is the midpoint of a distribution
  - Defined by number of scores
  - Often is not the balance point of the scores
- Both measure central tendency, using two different concepts of "middle"

3.4 The Mode

- The mode is the score or category that has the greatest frequency of any score in the frequency distribution
  - Can be used with any scale of measurement
  - Corresponds to an actual score in the data
- It is possible to have more than one mode

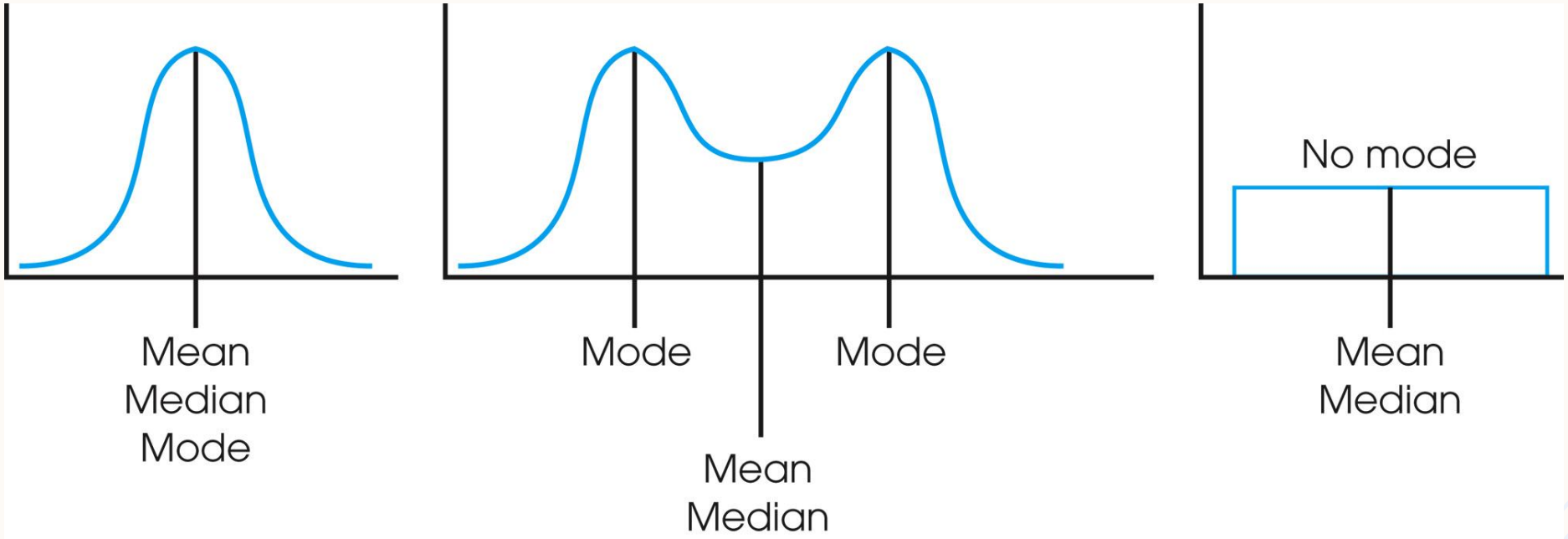
Figure 3.6
Bimodal Distribution



3.5 Central Tendency and the Shape of the Distribution

- Symmetrical distributions
  - Mean and median have same value
  - If exactly one mode, it has same value as the mean and the median
  - Distribution may have more than one mode,
    or no mode at all

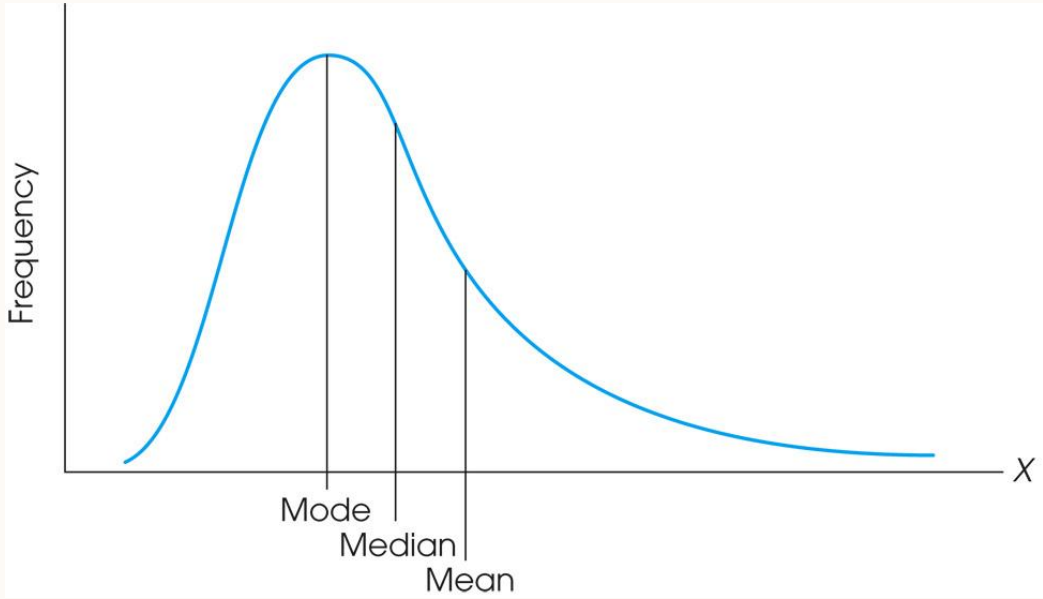
Figure 3.10 Symmetrical Distributions

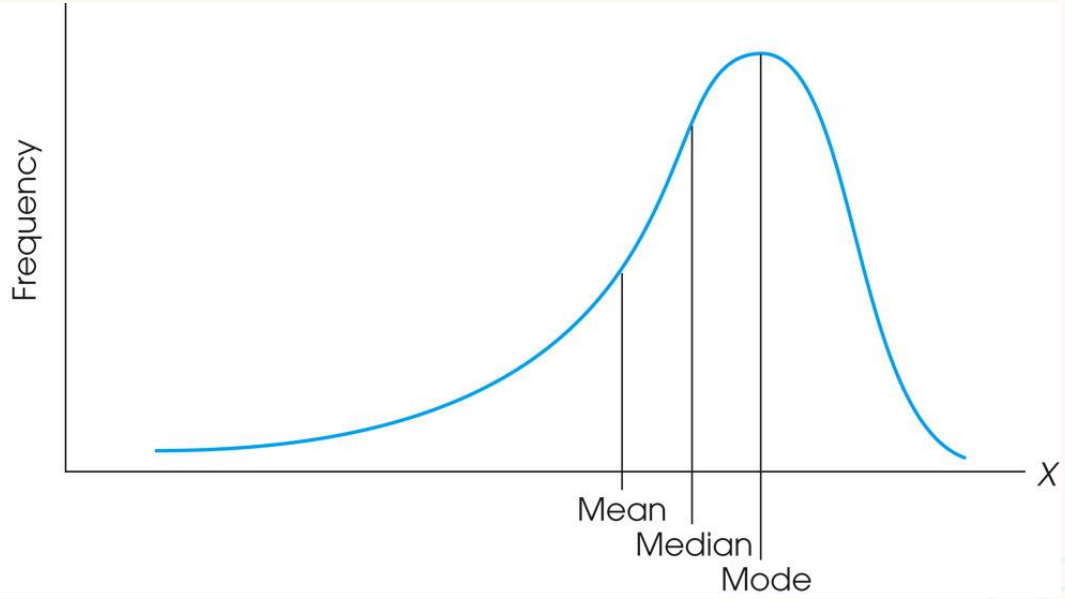


Central Tendency in Skewed Distributions

- Mean, influenced by extreme scores, is found far toward the long tail (positive or negative)
- Median, in order to divide scores in half, is found toward the long tail, but not as far as the mean
- Mode is found near the short tail.
- If Mean – Median > 0, the distribution is positively skewed.
- If Mean – Median < 0, the distribution is negatively skewed

Skewed Distributions





3.6 Selecting a Measure of Central Tendency

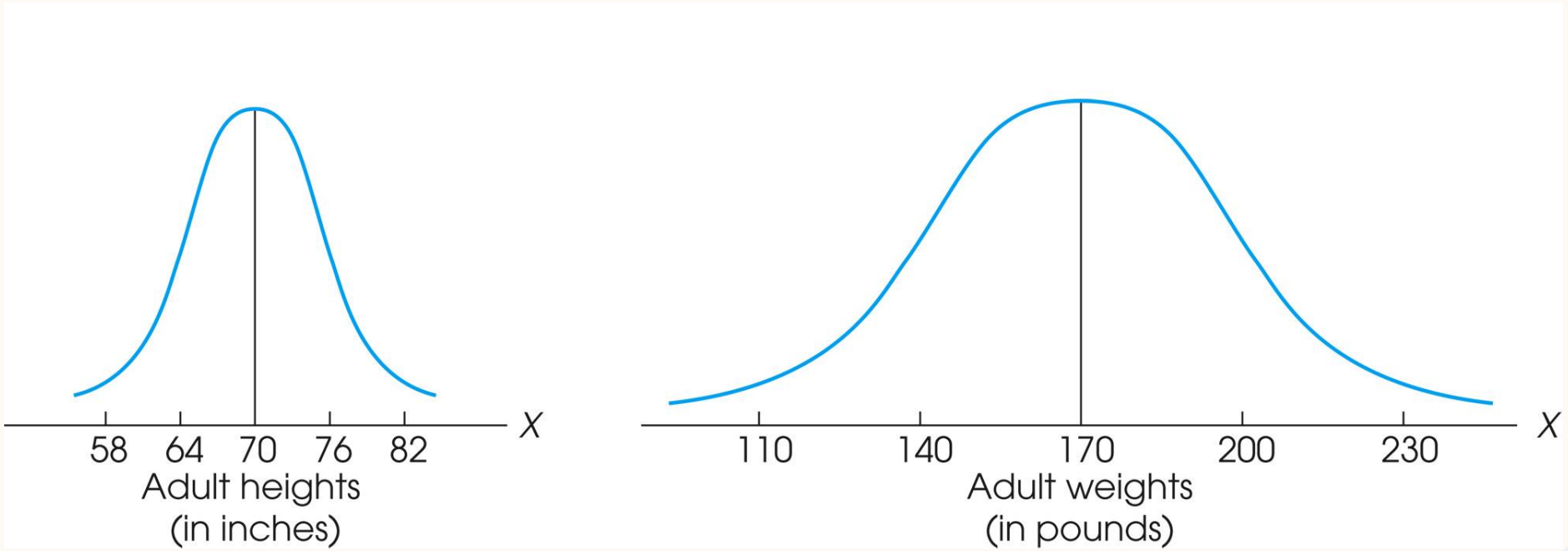
Measure of Central Tendency	Appropriate to choose if …	Could be misleading if…
Mean	• You can calculate ∑X • You know the value of every score	•Extreme scores •Skewed distribution •Undetermined values •Open-ended distribution •Ordinal scale •Nominal scale
Median	•Extreme scores •Skewed distribution •Undetermined values •Open-ended distribution •Ordinal scale	•Nominal scale
Mode	•Nominal scales •Discrete variables •Describing shape	•Interval or ratio data, except to accompany mean or median

Variability

4.1 Overview

- Variability can be defined several ways
  - A quantitative *distance* measure based on the differences between scores
  - Describes *distance* of the spread of scores or *distance* of a score from the mean
- Purposes of Measure of Variability
  - Describe the distribution
  - Measure how well an individual score represents the distribution

Population Distributions



Three Measures of Variability

- The Range
- The Variance
- The Standard Deviation

4.2 The Range

- The distance covered by the scores in a distribution
  - From smallest value to highest value
- For continuous data, real limits are used

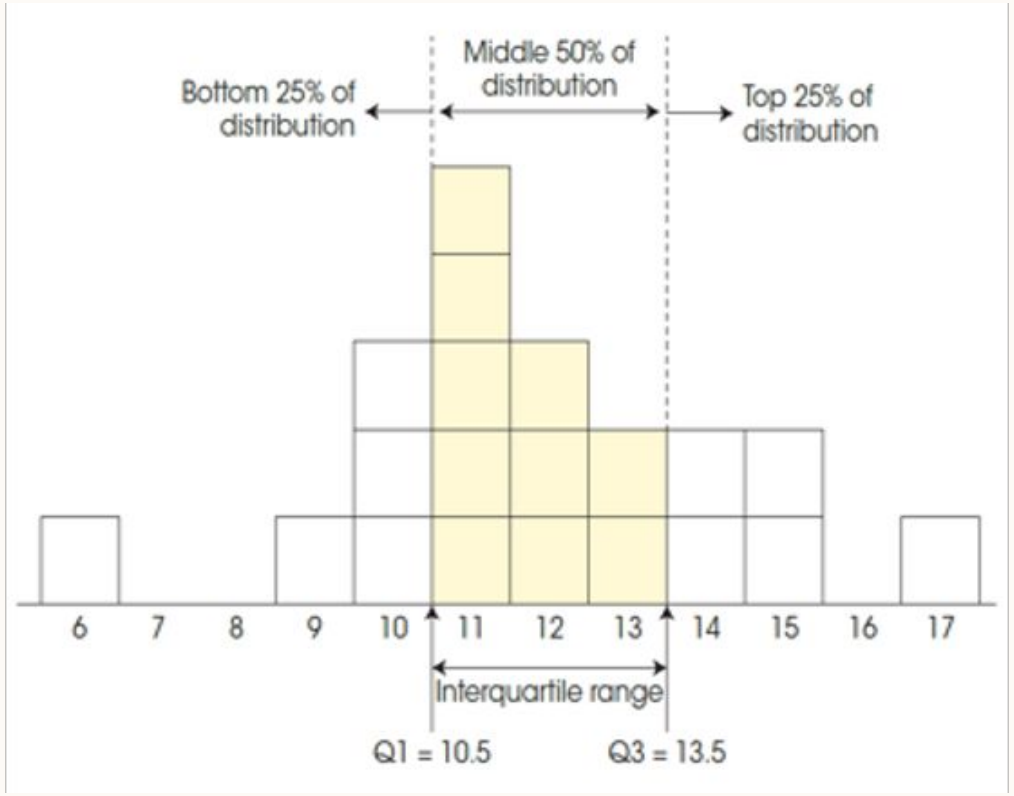
$$\text{range} = \text{URL for } X_{max} - \textit{LRL for } X_{min}$$

- Based on two scores, not all the data
  - An imprecise, unreliable measure of variability

The Interquartile Range

- Quartiles are the scores having percentile ranks of 25%, 50%, 75%, and 100%
  - These are called the first, second, thirds and fourth quartiles, respectively
  - Divide the distribution in four equal parts that each corresponds to 25% of the distribution
- The interquartile range (IQR) is the distance between the *X* values that correspond to the first (Q1) and third quartiles (Q3)
  - Scores fall in the middle 50% of the distribution

Figure 4.1
The Interquartile Range



4.3 Standard Deviation and Variance for a Population

- Most common and most important measure of variability is the standard deviation
  - A measure of the standard, or average, distance from the mean
  - Describes whether the scores are clustered closely around the mean or are widely scattered
- Calculation differs for population and samples
- Variance is a necessary ***companion concept*** to standard deviation but ***not*** the same concept

Defining the Standard Deviation

- Step One: Determine the Deviation
- Deviation is distance from the mean

$$\text{Deviation score} = X - \mu$$

- Step Two: Find a "sum of deviations" to use as a basis of finding an "average deviation"
  - Two problems
    - Deviations sum to 0 (because M is balance point)
    - If sum always 0, "Mean Deviation" will always be 0.
  - Need a new strategy!

Defining the Standard Deviation (continued)

- Step Two *Revised*: Remove negative deviations
  - ***First*** square each deviation score
  - Then sum the *Squared* Deviations (SS)
- Step Three: Average the *squared* deviations
  - Mean Squared Deviation is known as "***Variance***"
  - Variability is now measured in squared units

Population variance equals mean (average) squared deviation (distance) of the scores from the population mean

Defining the Standard Deviation (continued)

- Step Four:
  - Goal: to compute a measure of the "standard" (average) distance of the scores from the mean
  - Variance measures the average *squared* distance from the mean—not quite our goal
- Adjust for having squared all the differences by taking the square root of the variance
- $Standard\ Deviation = \sqrt{Variance}$

Figure 4.2

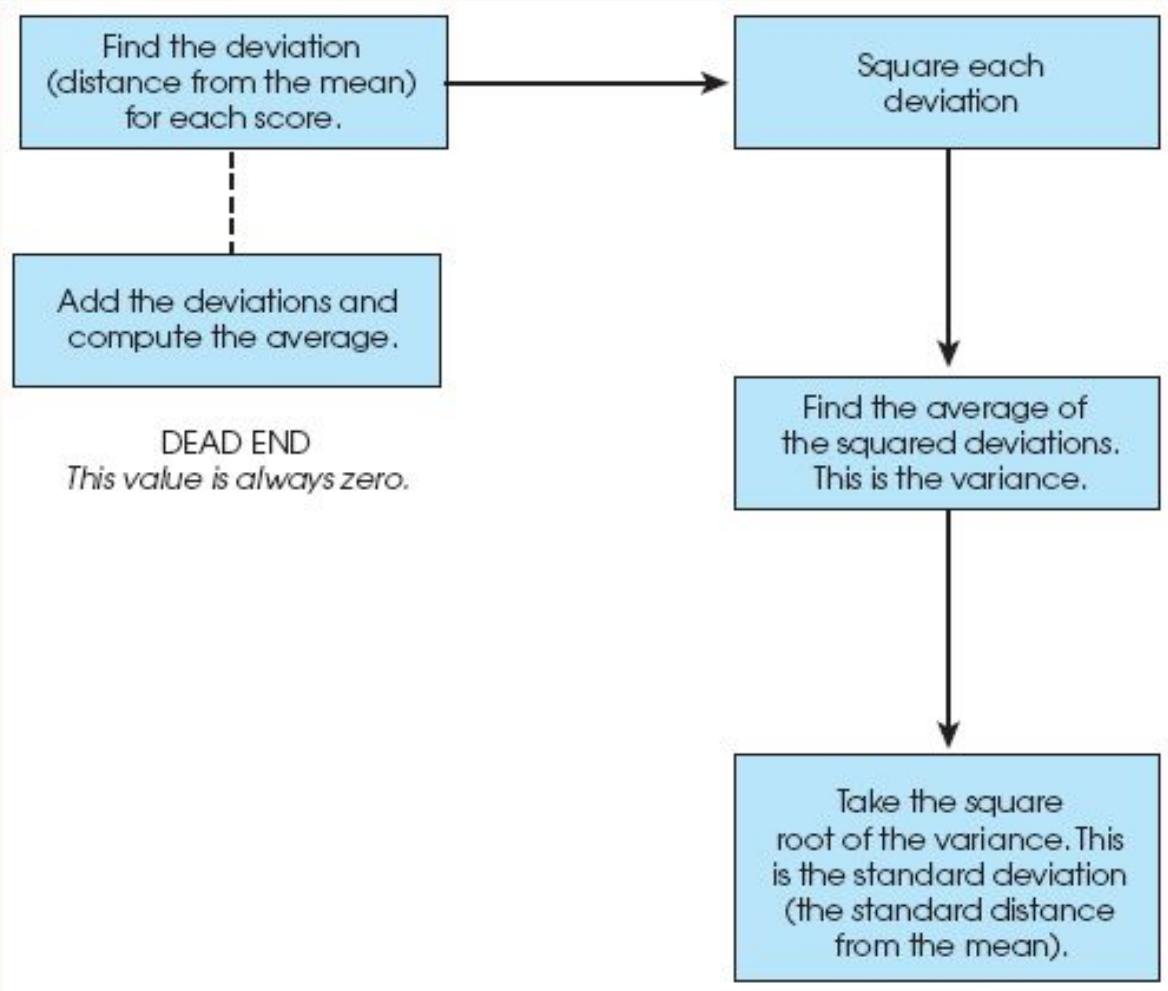
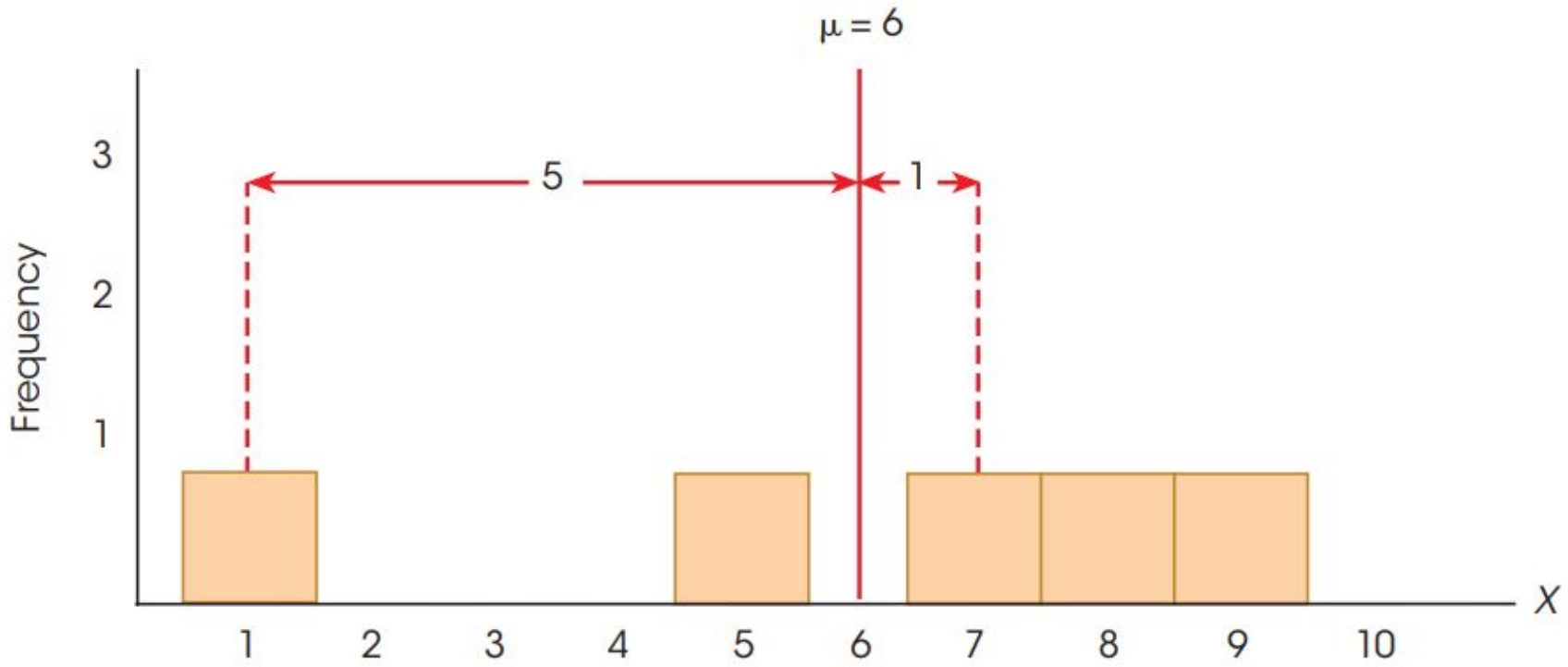

Find the deviation (distance from the mean) for each score.
Square each deviation
Add the deviations and compute the average.
DEAD END
This value is always zero.
Find the average of the squared deviations. This is the variance.
Take the square root of the variance. This is the standard deviation (the standard distance from the mean).

Figure 4.3

FIGURE 4.3
A frequency distribution histogram for a population of $N = 5$ scores. The mean for this population is $\mu = 6$. The smallest distance from the mean is 1 point and the largest distance is 5 points. The standard distance (or standard deviation) should be between 1 and 5 points.



Population Variance Formula

$$\text{Variance} = \frac{\text{sum of squared deviations}}{\text{number of scores}}$$

- SS (sum of squares) is the sum of the squared deviations of scores from the mean

- Two formulas for computing SS

Example

We will calculate the variance and standard deviation for the following population of $N = 5$ scores:

1, 9, 5, 8, 7

Remember that the purpose of standard deviation is to measure the standard distance from the mean, so we begin by computing the population mean. These five scores add up to $\Sigma X = 30$ so the mean is $\mu = \frac{30}{5} = 6$. Next, we find the deviation (distance from the mean) for each score and then square the deviations. Using the population mean $\mu = 6$, these calculations are shown in the following table.

Score X	Deviation $X - \mu$	Squared Deviation $(X - \mu)^2$	
1	−5	25	
9	3	9	
5	−1	1	
8	2	4	
7	1	1	
		40	= the sum of the squared deviations

For this set of $N = 5$ scores, the squared deviations add up to 40. The mean of the squared deviations, the variance, is $\frac{40}{5} = 8$, and the standard deviation is $\sqrt{8} = 2.83$. ■

Two formulas for SS

Definitional Formula

- Find each deviation score $(X-\mu)$
- Square each deviation score, $(X-\mu)^2$
- Sum up the squared deviations

$$SS = \sum (X - \mu)^2$$

Computational Formula

- Square each score and sum the squared scores
- Find the sum of scores, square it, divide by N
- Subtract the second part from the first

$$SS = \sum X^2 - \frac{\left(\sum X\right)^2}{N}$$

Example - Definitional Formula

We will compute SS for the following set of $N = 4$ scores. These scores have a sum of $\Sigma X = 8$, so the mean is $\mu = \frac{8}{4} = 2$. The following table shows the deviation and the squared deviation for each score. The sum of the squared deviation is $SS = 22$.

Score X	Deviation $X - \mu$	Squared Deviation $(X - \mu)^2$	
1	−1	1	$\Sigma X = 8$
0	−2	4	$\mu = 2$
6	+4	16	
1	−1	1	
		22	$\Sigma(X - \mu)^2 = 22$

Example - Computational Formula

X	X^2
1	1
0	0
6	36
1	1
$\Sigma X = 8$	$\Sigma X^2 = 38$

$$SS = \Sigma X^2 - \frac{(\Sigma X)^2}{N}$$

$$= 38 - \frac{(8)^2}{4}$$

$$= 38 - \frac{64}{4}$$

$$= 38 - 16$$

$$= 22$$

Caution Required!

When using the computational formula, remember…

$$SS \neq \sum X^2$$

$$SS \neq (\sum X)^2$$

$$\sum X^2 \neq (\sum X)^2$$

Population Variance: Formula and Notation

Formula

$$variance = \frac{SS}{N}$$

$$standard\ deviation = \sqrt{\frac{SS}{N}}$$

Notation

- Variance is the average of *squared* deviations, so we identify population variance with a lowercase Greek letter sigma squared: σ^2
- Standard deviation is the square root of the variance, so we identify it with a lowercase Greek letter sigma: σ

Learning Check

1. What is the value of *SS*, the sum of the squared deviations, for the following population of $N = 4$ scores? Scores: 1, 4, 6, 1
   a. 0
   b. 18
   c. 54
   d. $12^2 = 144$

2. Each of the following is the sum of the scores for a population of $N = 4$ scores. For which population would the definitional formula be a better choice than the computational formula for calculating *SS*.
   a. $\Sigma X = 9$
   b. $\Sigma X = 12$
   c. $\Sigma X = 15$
   d. $\Sigma X = 19$

3. What is the standard deviation for the following population of scores?
   Scores: 1, 3, 7, 4, 5
   a. 20
   b. 5
   c. 4
   d. 2

Sample Variance and Standard Deviation

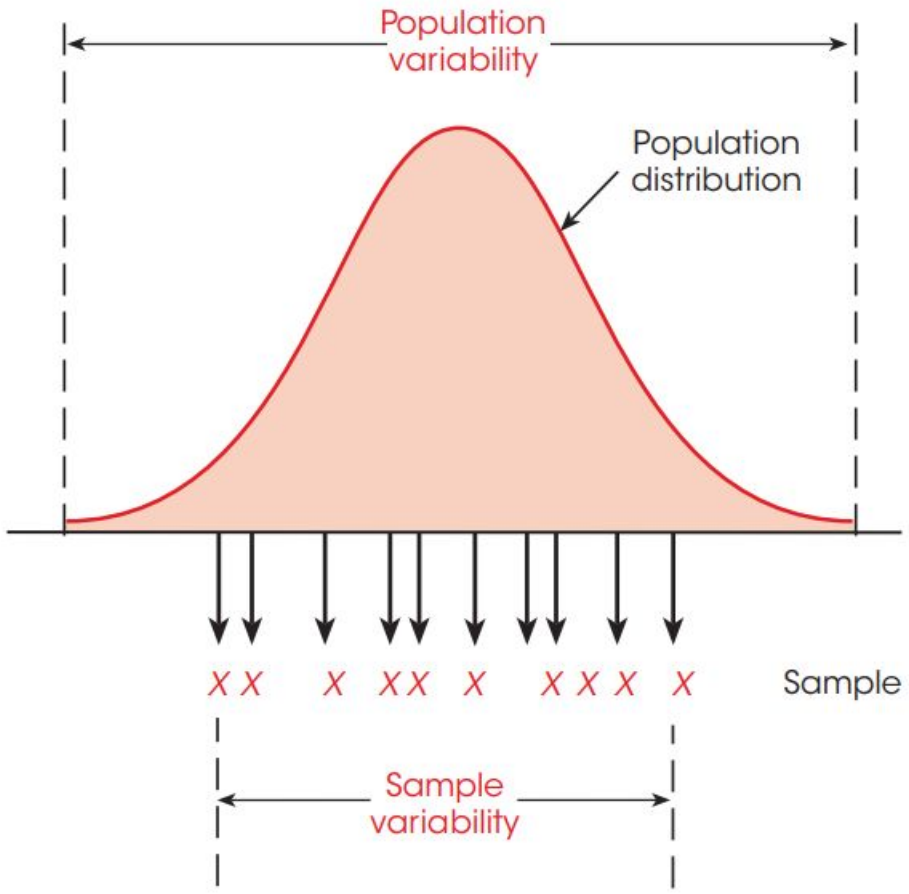


FIGURE 4.4
The population of adult heights forms a normal distribution. If you select a sample from this population, you are most likely to obtain individuals who are near average in height. As a result, the variability for the scores in the sample is smaller than the variability for the scores in the population.

Sample Variance and Standard Deviation

- Sum of Squares (SS) is computed as before
- Formula for Variance has n-1 rather than N in the denominator
- Notation uses *s* instead of σ

$$\text{variance of sample} = s^2 = \frac{SS}{n-1}$$

$$\text{standard deviation of sample} = s = \sqrt{\frac{SS}{n-1}}$$

Sample Variance and Standard Deviation

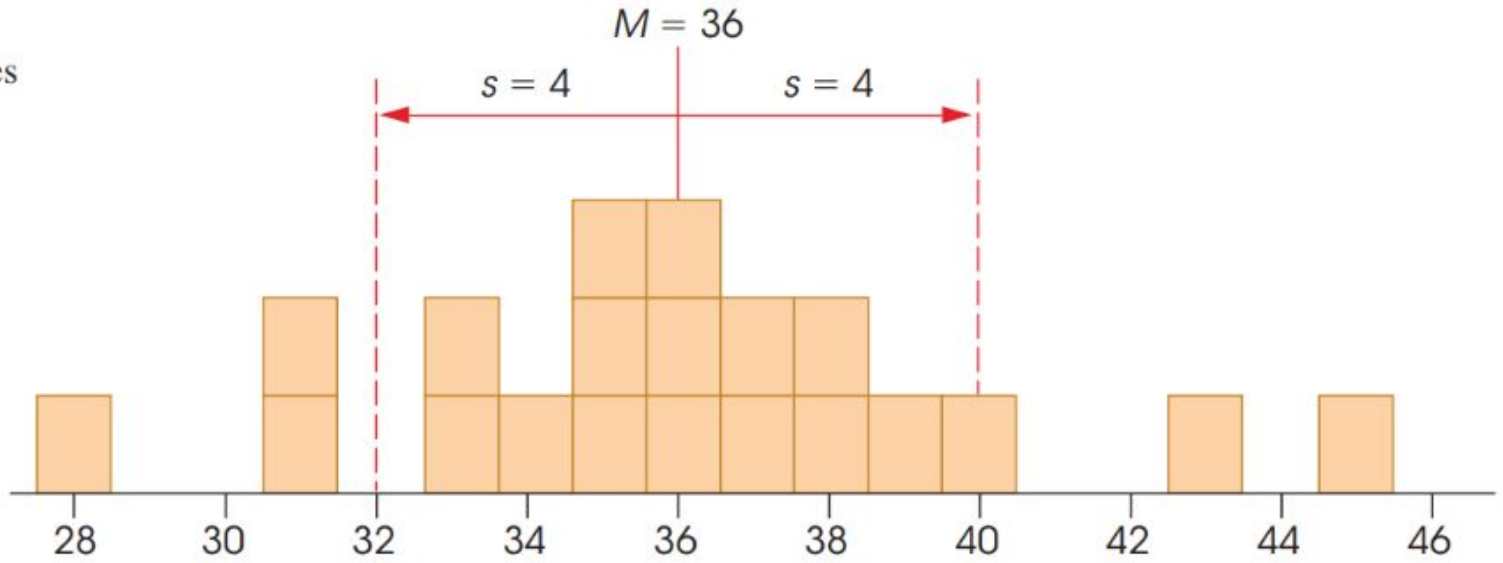
Notice that the sample formulas divide by n - 1 unlike the population formulas, which divide by N. This is the adjustment that is necessary to correct for the bias in sample variability. The effect of the adjustment is to increase the value you will obtain. Dividing by a smaller number (n - 1 instead of n) produces a larger result and makes sample variance an accurate and unbiased estimator of population variance.

Degrees of Freedom

- Population variance
  - Mean is known
  - Deviations are computed from a known mean
- Sample variance as estimate of population
  - Population mean is unknown
  - Using sample mean restricts variability
- Degrees of freedom
  - Number of scores in sample that are independent and free to vary
  - Degrees of freedom (df) = n – 1

Visualization of mean & standard deviation from a sample

FIGURE 4.7
A sample of $n = 20$ scores with a mean of $M = 36$ and a standard deviation of $s = 4$.

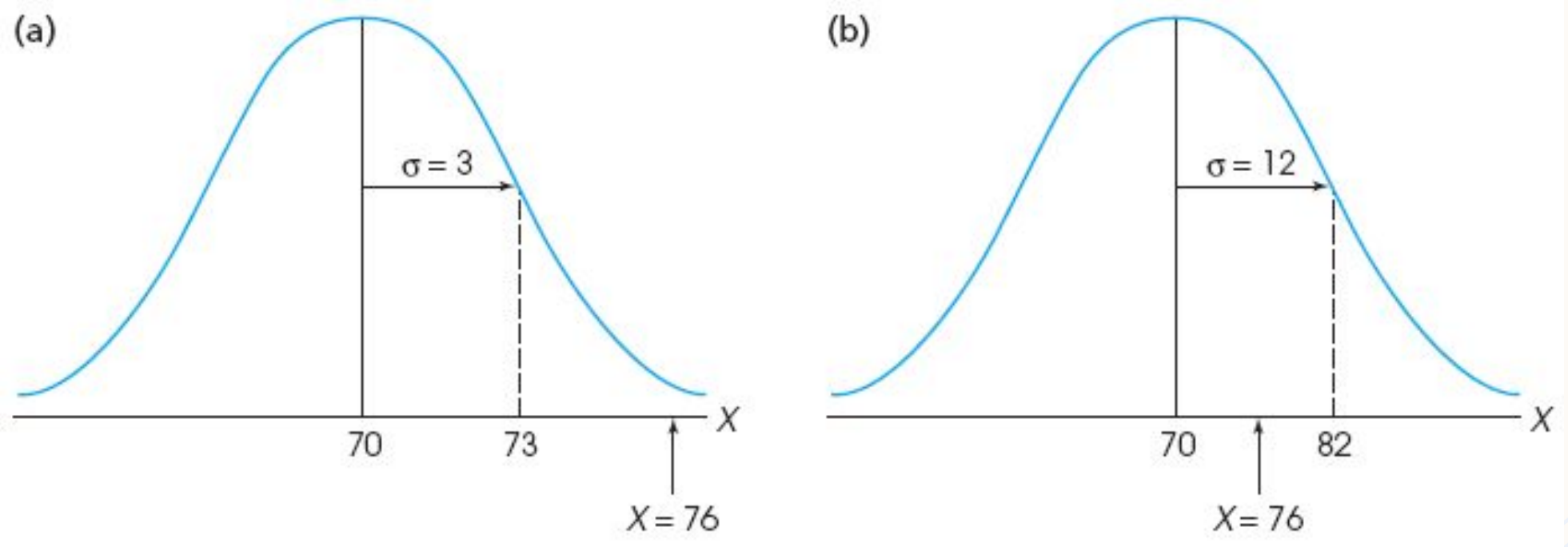


z-score

5.1 Purpose of *z*-Scores

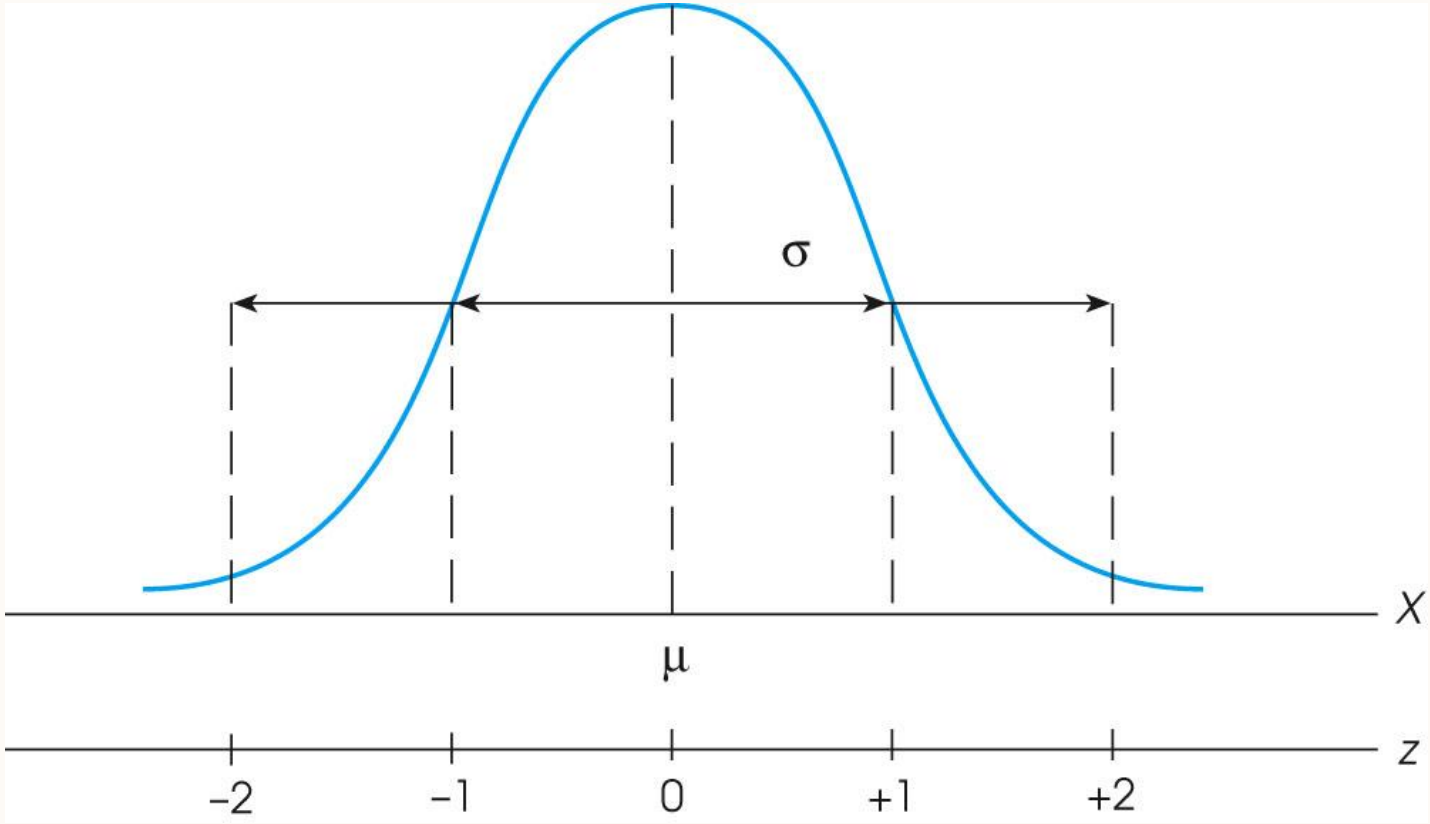
- Identify and describe location of every score in the distribution
- Standardize an entire distribution
- Take different distributions and make them equivalent and comparable

Two Exam Score Distributions



5.2 *z*-Scores and Location in a Distribution

- Exact location is described by *z*-score
  - Sign tells whether score is located above or below the mean
  - Number tells distance between score and mean in standard deviation units



Learning Check

1. If your exam score is $X = 60$, which set of parameters would give you the best grade?
   - **a.** $\mu = 65$ and $\sigma = 5$
   - **b.** $\mu = 65$ and $\sigma = 2$
   - **c.** $\mu = 70$ and $\sigma = 5$
   - **d.** $\mu = 70$ and $\sigma = 2$

2. For a distribution of exam scores with $\mu = 70$, which value for the standard deviation would give the highest grade to a score of $X = 75$?
   - **a.** $\sigma = 1$
   - **b.** $\sigma = 2$
   - **c.** $\sigma = 5$
   - **d.** $\sigma = 10$

3. Last week Sarah had a score of $X = 43$ on a Spanish exam and a score of $X = 75$ on an English exam. For which exam should Sarah expect the better grade?
   - **a.** Spanish
   - **b.** English
   - **c.** The two grades should be identical.
   - **d.** Impossible to determine without more information

Equation for *z*-Score

$$z = \frac{X - \mu}{\sigma}$$

- Numerator is a deviation score
- Denominator expresses deviation in standard deviation units

Determining a Raw Score from a *z*-Score

- $z = \dfrac{X - \mu}{\sigma}$ so $X = \mu + z\sigma$

- Algebraically solve for X to reveal that…
- Raw score is simply the population mean plus (or minus if *z* is below the mean) *z* multiplied by population the standard deviation

Computing *z*-Scores for a Sample

- Populations are most common context for computing *z*-scores
- It is possible to compute *z*-scores for samples
  - Indicates relative position of score in sample
  - Indicates distance from sample mean
- Sample distribution can be transformed into *z*-scores
  - Same shape as original distribution
  - Same mean *M* and standard deviation *s*

Learning Check

1. A population of scores has $\mu = 44$. In this population, an X value of 40 corresponds to $z = -0.50$. What is the population standard deviation?

a. 2

b. 4

c. 6

d. 8

2. A population of scores has $\sigma = 4$. In this population, an X value of 58 corresponds to $z = 2.00$. What is the population mean?

a. 54

b. 50

c. 62

d. 66

3. A population of scores has $\sigma = 10$. In this population, a score of $X = 60$ corresponds to $z = -1.50$. What is the population mean?

a. -30

b. 45

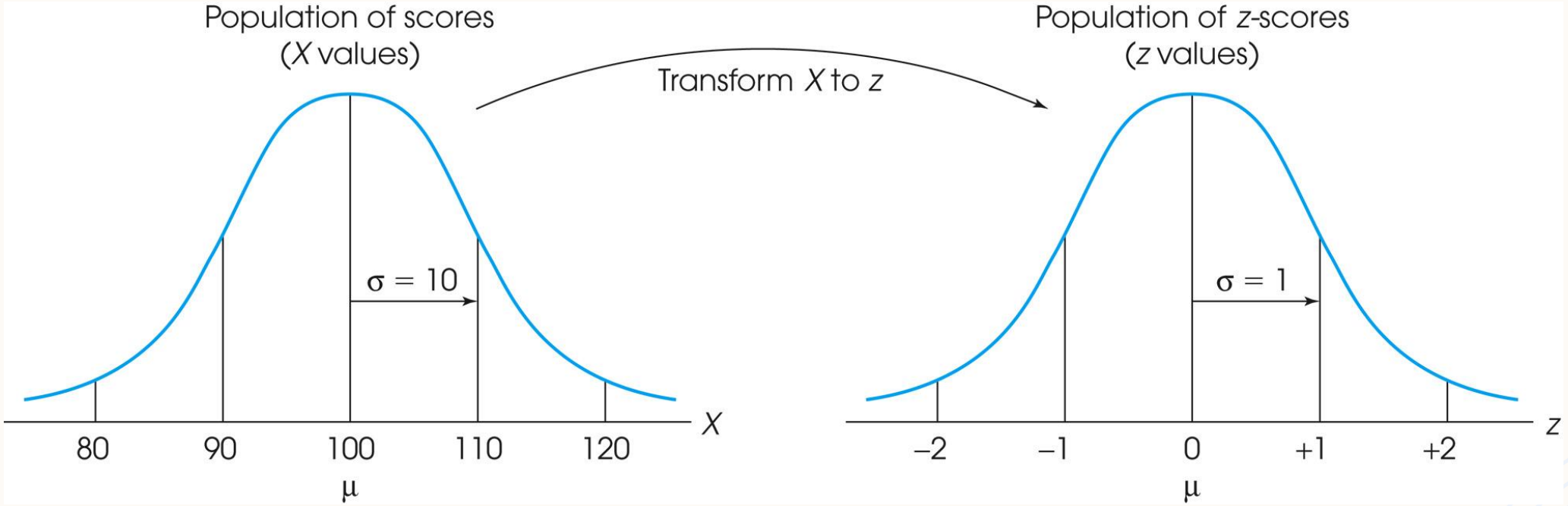
c. 75

d. 90

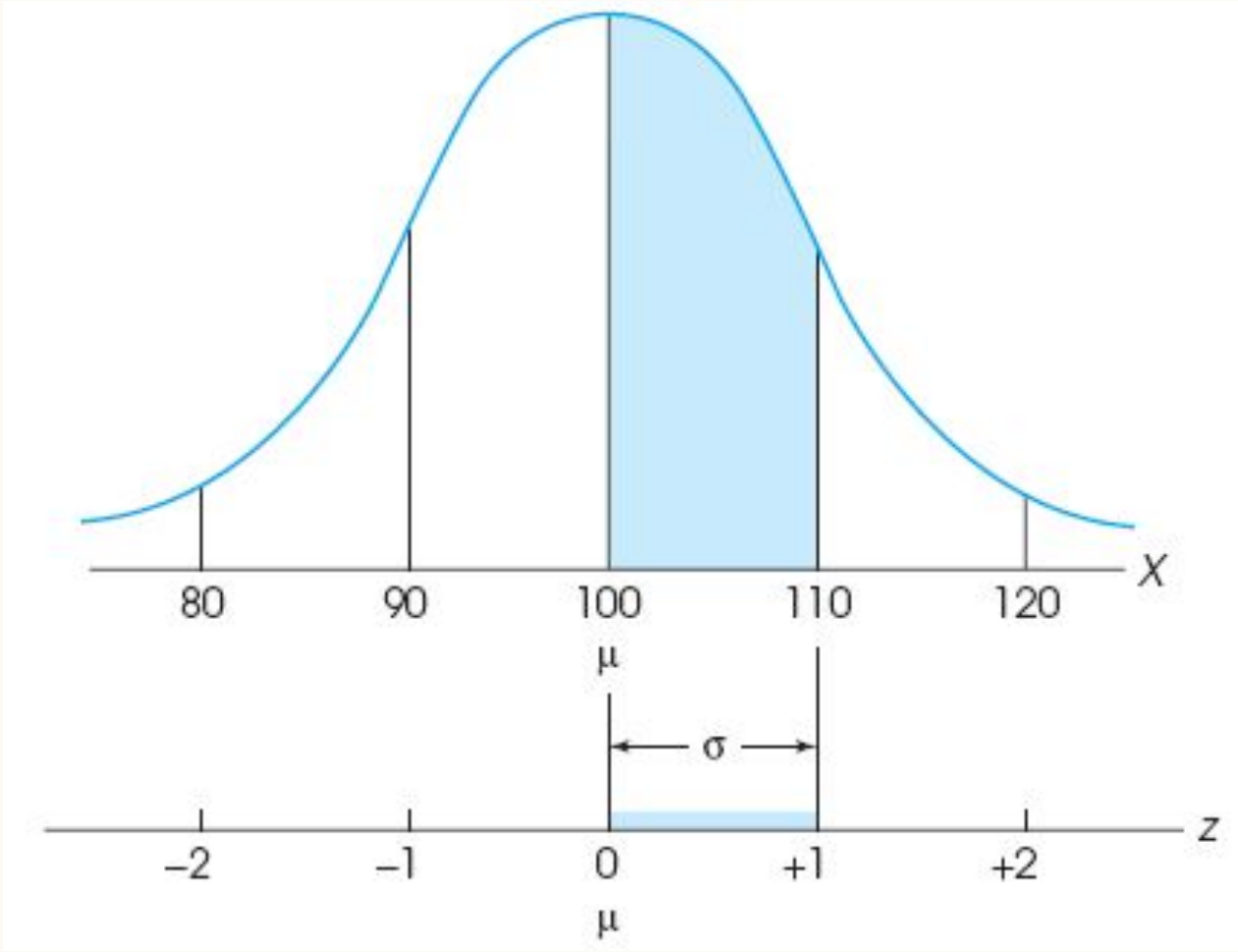
5.3 Standardizing a Distribution

- Every X value can be transformed to a z-score
- Characteristics of z-score transformation
  - Same shape as original distribution
  - Mean of z-score distribution is always 0.
  - Standard deviation is always 1.00
- A z-score distribution is called a standardized distribution

Transforming a Population of Scores



Axis Re-labeling After *z*-Score Transformation



z-Scores Used for Comparisons

- All *z*-scores are comparable to each other
- Scores from different distributions can be converted to *z*-scores
- *z*-scores (standardized scores) allow the direct comparison of scores from two different distributions *because* they have been converted to the same scale

5.4 Other Standardized Distributions

- Process of standardization is widely used
  - SAT has $\mu = 500$ and $\sigma = 100$
  - IQ has $\mu = 100$ and $\sigma = 15$ Points
- Standardizing a distribution has two steps
  - Original raw scores transformed to z-scores
  - The z-scores are transformed to new X values so that the specific predetermined μ and σ are attained.

Creating a Standardized Distribution

EXAMPLE 5.9

An instructor gives an exam to a psychology class. For this exam, the distribution of raw scores has a mean of $\mu = 57$ with $\sigma = 14$. The instructor would like to simplify the distribution by transforming all scores into a new, standardized distribution with $\mu = 50$ and $\sigma = 10$. To demonstrate this process, we will consider what happens to two specific students: Maria, who has a raw score of $X = 64$ in the original distribution; and Joe, whose original raw score is $X = 43$.

STEP 1 Transform each of the original raw scores into z-scores. For Maria, $X = 64$, so her z-score is

$$z = \frac{X - \mu}{\sigma} = \frac{64 - 57}{14} = +0.5$$

For Joe, $X = 43$, and his z-score is

$$z = \frac{X - \mu}{\sigma} = \frac{43 - 57}{14} = -1.0$$

Remember: The values of μ and σ are for the distribution from which X was taken.

STEP 2 Change each z-score into an X value in the new standardized distribution that has a mean of $\mu = 50$ and a standard deviation of $\sigma = 10$.

Maria's z-score, $z = +0.50$, indicates that she is located above the mean by $\frac{1}{2}$ standard deviation. In the new, standardized distribution, this location corresponds to $X = 55$ (above the mean by 5 points).

Joe's z-score, $z = -1.00$, indicates that he is located below the mean by exactly 1 standard deviation. In the new distribution, this location corresponds to $X = 40$ (below the mean by 10 points).

The results of this two-step transformation process are summarized in Table 5.1. Note that Joe, for example, has exactly the same z-score ($z = -1.00$) in both the original distribution and the new standardized distribution. This means that Joe's position relative to the other students in the class has not changed. ■

Figure 5.9 Creating a Standardized Distribution

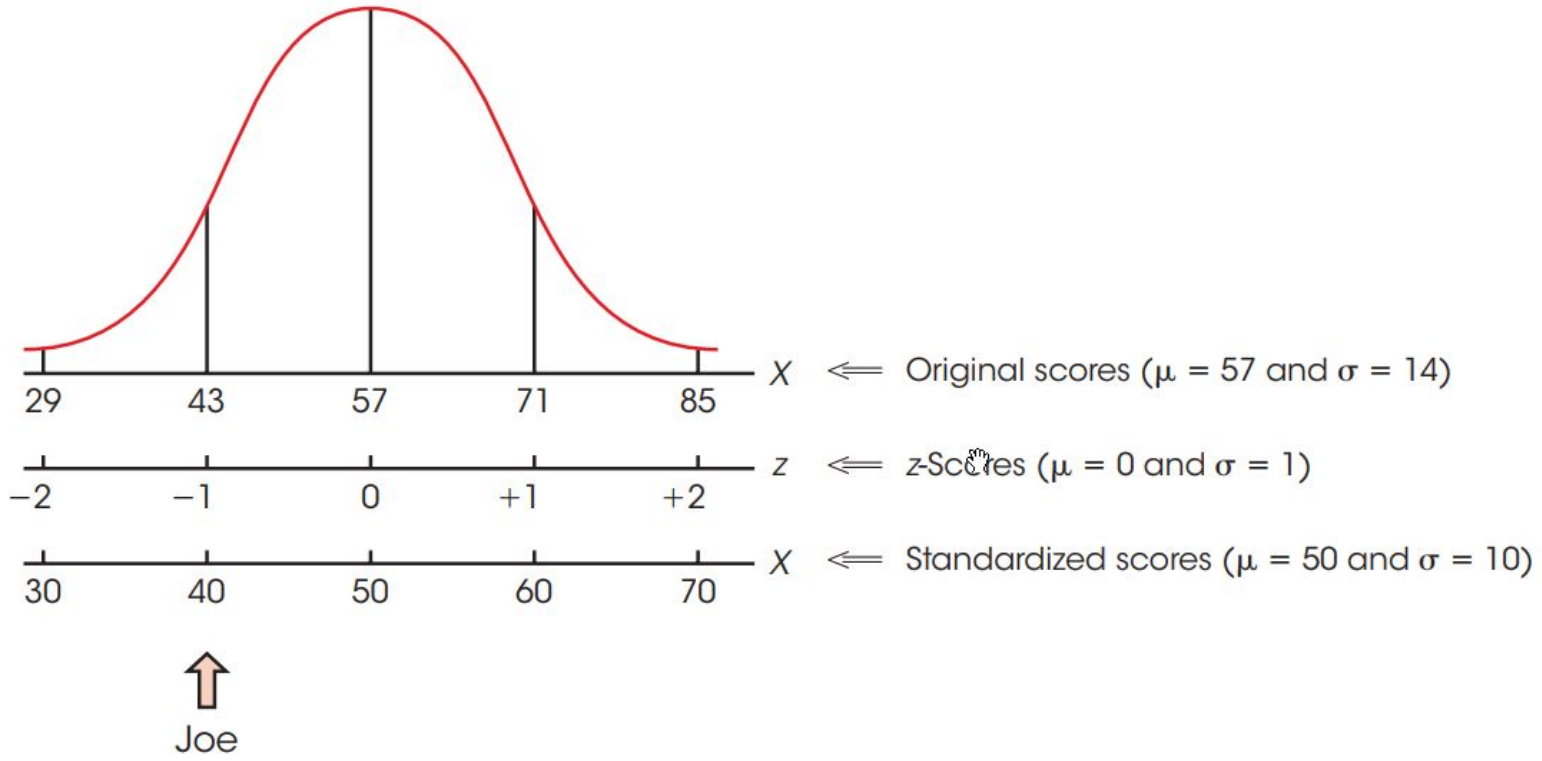


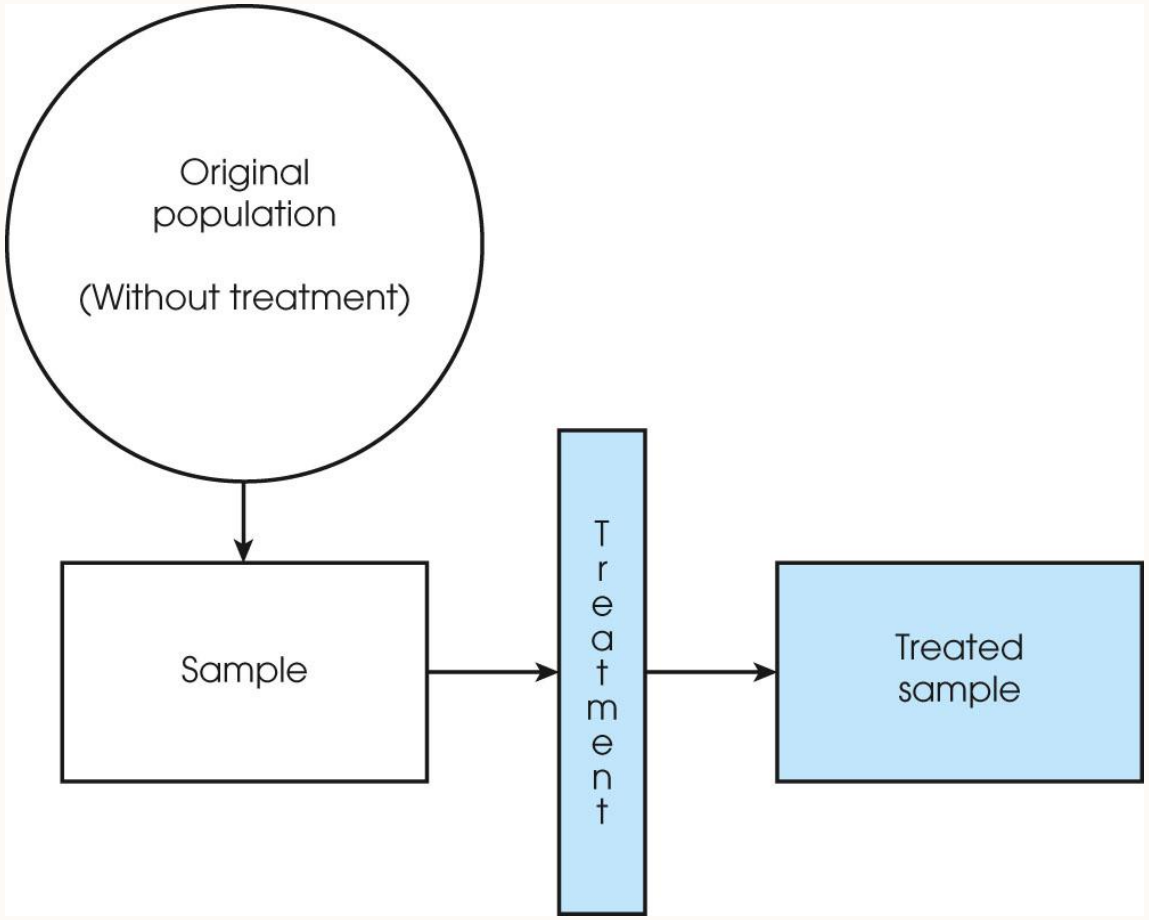
FIGURE 5.9

The distribution of exam scores from Example 5.9. The original distribution was standardized to produce a distribution with $\mu = 50$ and $\sigma = 10$. Note that each individual is identified by an original score, a z-score, and a new, standardized score. For example, Joe has an original score of 43, a z-score of -1.00, and a standardized score of 40.

Looking Ahead to Inferential Statistics

- Interpretation of research results depends on determining if (treated) a sample is "noticeably different" from the population
- One technique for defining "noticeably different" uses *z*-scores.

Conceptualizing the Research Study



THANK YOU